Shared Pell bounds for modular certificates

Explicit smaller Diophantine equations for ZFC consistency

Research prepared for Vladimir Reshetnikov's ProveIt project
AI-assisted derivation, implementation, and independent mathematical checks

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Abstract

We give a completely specified quartic polynomial in 241 nonnegative integer variables, with 189 quadratic residuals, 30,058 collected monomials, and coefficient height 85,187,728. Its no-zero statement is equivalent to Con(ZFC) by an ordinary arithmetic proof composed with the identified, retained C222–ZF–ZFC refutation calibration. The construction encodes finite proof certificates directly; it does not simulate K215 or instantiate Jones's universal polynomial.

The main change is to extend a single indexed Pell bound to a mixed family of exact-power and modular-numerator requests. The enlarged bound is defined before either kind of request is interpreted. A separate bootstrap uses the factorial ceiling equations themselves to justify a smaller digit radix. These changes allow one binomial coefficient, one indexed bound, and one congruence of squared index errors to serve the whole certificate. The resulting quartic improves all four measured size quantities of the incoming 261-variable direct quartic. A smaller-variable, denser companion and ten-positive-variable norm equations are also supplied.

Every principal residual is printed in the appendix. The archive includes the expanded polynomial, executable builders, restoration certificates, independently written checks, and source proofs. No consistency assumption is used to establish the equivalence. We claim neither a global simplicity minimum nor a newly completed end-to-end proof-assistant certificate.

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1 Results and the meaning of a simpler equation

The recommended equation in this packet is

$$\boxed{P_{241}(z_0,\dots,z_{240}) = \sum_{j=0}^{188} Q_j(\mathbf{z})^2 = 0, \qquad \mathbf{z} \in \mathbb{N}^{241}.} \tag{1}$$

Here and throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. The symbols Q_j are the literal quadratic polynomials in Appendix A, not unspecified representations selected by an existence theorem. The matching JSON file gives the same ordered rows and every coordinate; the expanded file gives every coefficient of their collected sum of squares.

Direct quartic over $\mathbb{N}$	Variables	Rows	Terms	Height
Incoming shared-binomial equation	319	254	10,283	86,842,335
Incoming shared-bound equation	261	194	31,679	85,566,955
Principal equation here	**241**	**189**	**30,058**	**85,187,728**
Dense companion here	239	187	387,494	269,805,528

All four equations have exact degree four. A "row" is one quadratic residual before the final sum of squares. "Terms" counts distinct nonzero monomials after collecting that single polynomial. Height is the largest absolute value of a collected coefficient. It is not the largest coefficient in a residual. The principal equation saves 20 variables, five residuals, 1,621 monomials, and 379,227 units of coefficient height relative to the 261-variable equation. These are reductions of about 7.66%, 2.58%, 5.12%, and 0.44%, respectively.

The dense companion lowers the variable count to 239, at a substantial cost in expanded support. It is included so that minimizing variables does not silently become minimizing every measure at once.

The comparison is against the complete incoming files at repository commit `f4d8db99e060339f3306a04d507c35bc1d0d68c5`, including all eight ZIPs then present in `docs/incoming`. The two principal parents are *Shared Binomial Certificates* and *Shared Pell Bound 261* [2, 3]. The repository already has equations with far fewer variables and very large fixed coefficients, including nine-positive constructions and a specialized 37-integer quartic. The results here advance the explicit direct-calculus, comparatively small-coefficient branch. They do not supersede those different tradeoffs.

This distinction is also standard in the literature: degree, number of unknowns, and the choice between $\mathbb{N}$ and $\mathbb{Z}$ are separate complexity parameters [6, 7]. None of the present numerical comparisons is a theorem that no better polynomial exists.

2 The precise logical statement

Fix the ordinary finite proof calculus for ZFC used by ProveIt's retained calibration. Let $\mathrm{Ref}_{\mathsf{ZFC}}(p)$ be its primitive-recursive predicate that p codes a finite refutation. Thus

$$\mathrm{Con}(\mathsf{ZFC}) \equiv \neg\exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathsf{ZFC}}(p).$$

The arithmetic argument of this packet, composed with the retained source calibration, proves

$$\mathsf{ZFC} \vdash \Big(\exists \mathbf{z} \in \mathbb{N}^{241}\ P_{241}(\mathbf{z}) = 0\Big) \longleftrightarrow \left(\exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathsf{ZFC}}(p)\right). \tag{2}$$

Taking negations yields exactly the requested consistency equivalence. The proof is uniform finite arithmetic, rather than an argument restricted to particular external numerals. Its ingredients are

induction on finite sequences, elementary Pell theory, integer divisibility, the Chinese remainder theorem, and explicit finite polynomial substitution. They can be carried out on the natural numbers of a model of ZFC.

There are three different levels of evidence, and they should be kept separate. First, the literal polynomial and its exact metrics are checked by finite computation. Second, the new transformations have ordinary proofs of soundness and constructive completeness. Third, the endpoint from the numerical C222 calculus to ZFC refutations is imported from the identified repository calibration, whose source texts and pins are retained. This work does not newly re-prove every transitive calibration dependency or construct a complete Lean/Rocq proof term for (2).

The imported endpoint is the ordinary argument associated with `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. We use that calculus endpoint before any machine-specific endpoint. Neither the builder nor the definition of the direct quartic reads the K215 transition table. The numerical coefficients in (1) are fixed by the supplied finite code and data; their definition does not branch on whether ZFC is consistent.

Remark 2.1 (Unrestricted integer interpretation). The principal displayed domain is $\mathbb{N}$, not all of $\mathbb{Z}$. If an equation quantified over unrestricted integers is required, there is an explicit quartic conversion without any appeal to a universal polynomial. Retain each z_i as an integer variable and add three integer variables a_i, b_i, c_i. Set

$$\widehat{P}(\mathbf{z},\mathbf{a},\mathbf{b},\mathbf{c}) = P_{241}(\mathbf{z}) + \sum_{i=0}^{240}(8z_i+1-a_i^2-b_i^2-c_i^2)^2.$$

Each new row forces $z_i \geq 0$. Conversely every $8z_i+1$, $z_i \geq 0$, is a sum of three integer squares by the three-squares theorem. Thus $\widehat{P}=0$ has an integer solution exactly when (1) has a natural solution. This is a transparent 964-variable quartic conversion, not a claimed improvement on the repository's specialized unrestricted-integer equations.

3 The direct proof-certificate source

The inherited source uses finite multisets of numerical proof rules. Its truth label is 0 and its contradiction label is 1. Formula constructors are ordinary polynomials; for example the Cantor-style implication constructor is

$$I(a,b) = 2(a+b)(a+b+1)+4b.$$

Membership, universal closure, and equality are the consecutive tags $I(a,b)+1$, $I(a,b)+2$, and $I(a,b)+3$. The precise nine axiom families, modus ponens, generalization, and guarded bundle are retained literally in the source compiler. This packet changes the arithmetic certificates used to validate the bounded source, not the calculus.

Choose a horizon $t \geq 12$ and an integer F later forced to be $t!$. Attach weight $1+jF$ to a positive formula label $j \leq t$ and weight 1 to label 0. For a rule with premises p_1, p_2 and conclusion c, a pending-product accumulator satisfies

$$S'(1+Fc) = S(1+Fp_1)(1+Fp_2). \tag{3}$$

The endpoints are $S_0 = 1+F$ and $S_t = 1$. A bounded sequence of such relations is encoded using explicit beta remainders.

When $F = t!$, weights attached to distinct positive labels at most t are pairwise coprime. A prime common to $1+jF$ and $1+kF$ would divide $j-k$, hence be at most t and divide F, contradicting its divisibility of $1+jF$. Telescoping (3) gives a product identity. Prime valuations recover exact multiplicity balance: there is one excess conclusion of label 1 and zero excess for every other positive label.

This balance entails derivability even if the selected rules contain cycles. Let D be the closure of the axioms and truth under the finite selected rules. If label 1 were outside D, every selected rule whose conclusion lies outside D would have at least one premise outside D. Counting those premises would contradict the excess of one conclusion outside D. Thus an accepted flow yields a finite C222 refutation.

Conversely, expand a finite proof DAG into a proof tree, choose t above all labels, parameters, tree size, and prescribed local polynomial bounds, and pad with truth rules. Traversing the pending proof multiset backwards constructs the positive accumulator products. After t is fixed, set $F = t!$, choose a sufficiently large multiple y of F, and use CRT to obtain the beta code x. This ordering matters: the large accumulator values do not have to be bounded by t.

The local source has eight encoded columns and one encoded index. Its local polynomial L expresses the rule disjunction and beta equations; its additional polynomial J bounds all possible rule labels. Their literal definitions, finite majorant certificates, and exact source bindings are included. Their fixed bounds used below are

$$E_0 = 2^{23}, \qquad \ell_0 = 2^{22}.$$

These constants concern polynomial growth bounds. They are not hidden free parameters of (1).

4 One indexed Pell bound for exact and modular requests

For an integer $A \geq 2$, write

$$\begin{aligned} X_0(A) &= 1, & Y_0(A) &= 0, \\ X_{n+1}(A) &= AX_n(A) + (A^2-1)Y_n(A), & Y_{n+1}(A) &= X_n(A) + AY_n(A). \end{aligned}$$

Then $X_n(A)^2 - (A^2-1)Y_n(A)^2 = 1$. All nonnegative integer solutions of this Pell equation are obtained uniquely in this way. This follows by the usual inverse descent; it is also a retained input to the parents' explicit indexed-Pell graphs. The ordinates $Y_n(A)$ strictly increase, and

$$X_n(A) \equiv 1, \qquad Y_n(A) \equiv n \pmod{A-1}. \tag{4}$$

4.1 The essential change to the bound

There are two kinds of requests. An exact-power request has a base b, an exponent k, and an output m, intended to mean $m = b^k$. A modular request has an upper index r, intended to supply the residue of $(q+1)^r$ for a binomial digit. A bound including only the exact-power triples is insufficient: a modular upper index can be arbitrarily larger than those exponents.

Define $\mathcal{A}$ to be the set of distinct actual natural-valued atoms in all exact-power triples, together with every modular upper index. The new frame is the literal polynomial equation

$$W = 2 + \sum_{a \in \mathcal{A}} a. \tag{5}$$

Every independent positive seed index k_i belongs to $\mathcal{A}$. If there are m such distinct seeds, then

$$0 < k_i < W, \qquad m \leq \sum_i k_i \leq W - 2.$$

These facts hold before any power or binomial has been interpreted. In the selected construction positivity comes from the raw horizon, coordinate shifts, and modulus rows. Indices supplied by an exact Pell addition formula are recorded as derived indices rather than unchecked extra seeds.

To make the Pell base sufficiently large, use natural z, L, σ, A with

$$L^2 - ((W+1)^2 - 1)(W(z+1))^2 = 1, \qquad A = L + \sigma. \tag{6}$$

Pell classification gives $L = X_e(W+1)$ and $W(z+1) = Y_e(W+1)$ for a positive e. Modulo W, the latter is congruent to e, so $W \mid e$ and $e \geq W$. Therefore

$$A \geq L \geq (W+1)^W, \qquad A - 1 > W^3 \quad (W \geq 3). \tag{7}$$

Conversely, $e = W$ supplies L, z, and every larger A remains available through σ. This freedom is essential for the later root congruence.

4.2 The simultaneous index theorem

Theorem 4.1 (Shared bound and one error congruence)**.** *Let $W \geq 3$, $A - 1 > W^3$, $0 < k_i < W$, and $m \leq W - 2$. Suppose $V = Y_W(A)$ is certified once. The following equations in nonnegative integers force every principal pair to have exactly its requested index:*

$$x_i^2 - (A^2 - 1)y_i^2 = 1 \qquad (1 \leq i \leq m), \tag{8}$$

$$V = W + 1 + s + \sum_{i=1}^{m} y_i, \tag{9}$$

$$\sum_{i=1}^{m} (y_i - k_i)^2 = (A-1)u. \tag{10}$$

Specifically, $x_i = X_{k_i}(A)$ and $y_i = Y_{k_i}(A)$. Conversely the canonical pairs extend to a solution of these equations.

Proof. Write $y_i = Y_{j_i}(A)$ using Pell classification, initially permitting $j_i = 0$. Equation (9) gives $y_i < V$, hence $0 \leq j_i < W$. Reduce (10) modulo $A - 1$ using (4):

$$A - 1 \mid E := \sum_i (j_i - k_i)^2.$$

But

$$0 \leq E \leq m(W-1)^2 \leq (W-2)(W-1)^2 < W^3 < A - 1.$$

Consequently $E = 0$, and every $j_i = k_i$. The potentially enormous ordinates y_i were not assumed to be small residues; the bound applies to their decoded indices.

For completeness, $Y_{k_i}(A) \geq k_i$ and their differences are multiples of $A - 1$, so the quotient in (10) is a nonnegative integer. Also $y_i \leq Y_{W-1}(A)$ and the recurrence gives $Y_W(A) > (2A-1)Y_{W-1}(A)$. Since $m \leq W - 2$ and $A > W + 1$, $Y_W(A) - \sum_i y_i > W$; hence the slack in (9) is natural. □

The theorem is inherited in its original exact-power setting from the 261-variable packet. The new application is its extension to the mixed family, using the enlarged (5). Omitting the modular upper indices would invalidate the size argument and is explicitly detected by the independent checker.

4.3 The bound itself is a polynomial certificate

The notation $V = Y_W(A)$ is proof notation, not an arithmetic function left in the final equation. Retain one complete indexed-Pell certificate:

$$\begin{aligned} U^2 - (A^2-1)V^2 &= 1, & H^2 - (A^2-1)J^2 &= 1, \\ S^2 - (B_c^2-1)R_c^2 &= 1, & B_c &= 1 + 4V(d+1), \\ J &= V^2(e+1), & B_c &= A + Hf, \\ S &= U + Hg, & R_c &= W + 4Vh. \end{aligned}$$

All letters here are natural; intermediate products are compiled into quadratic graph rows. The hypothesis $W \le V$ of the retained indexed Pell lemma follows from (9). Its soundness gives $U = X_W(A)$ and $V = Y_W(A)$, and its constructive completeness supplies the comparison and congruence witnesses. The private witnesses of this one anchor are not inserted into $\mathcal{A}$: doing so would introduce an unnecessary circular definition.

5 A smaller radix justified before factorial semantics

The shared-binomial parent already identifies the same binomial coefficient as both a factorial certificate and the reciprocal CRT product. The new radix uses this sharing together with the raw ceiling rows.

Let $t = t_0 + 12$ and take the literal source quantities

$$B = b_0 + 2, \quad R = 2 + t + x + y + F, \quad K = R, \quad N = B^K, \quad V_0 = (F^2+1)^t.$$

Let S_0 be the sum of the nine positive CRT representatives. Define

$$q = N(S_0+1)^K. \tag{11}$$

The displayed powers are realized by the simultaneous polynomial power graph; they are not variable exponents in the final polynomial. Compared with the earlier self-sized radix, the term V_0 has disappeared from its base. The aggregate upper bound on the nine representatives is also unnecessary: their actual sum sets the radix after they are chosen.

The retained ceiling rows are

$$V_0 + B = BF + r + 1, \qquad B = r + s + 1. \tag{12}$$

Thus $V_0 + s = BF$ and $0 \le s < B$. Once the exact powers are decoded, $V_0 \ge 1$, so $F \ge 1$. Since $t \ge 12$,

$$F^2 < V_0 \le BF \quad \Longrightarrow \quad F < B, \qquad V_0 < B^2 \le B^K = N \le q. \tag{13}$$

This uses neither $F = t!$ nor the interpretation of B as a binomial. The raw inequality $V_0 < B^2$ already occurs in the incoming unified-binomial packet [5]; its application to the new self-sized radix and mixed-index certificate is the combination used here.

Lemma 5.1 (Uniform lower binomial digits)**.** *For integers $r \ge 1$, $t \ge 1$, and $0 \le j \le t$,*

$$\binom{r+t}{j} \le (r+1)^t.$$

Proof. Induct on t. The case $t = 1$ is immediate. If $j = t$, use $\binom{r+t}{t} = \prod_{i=1}^{t}(1 + r/i) \le (r+1)^t$. For $1 \le j < t$, Pascal's identity and induction bound its two terms by $2(r+1)^{t-1} \le (r+1)^t$. The case $j = 0$ is immediate. □

Apply the lemma with $r = F^2$. All digits through place t in $(q+1)^{F^2+t}$ are at most $V_0 < q$. The single exact-binomial digit certificate therefore forces

$$B = \binom{F^2+t}{t}. \tag{14}$$

No later CRT range estimate is needed to recognize this first binomial.

Lemma 5.2 (Squared-radius factorial fixed point)**.** *For $t \geq 6$ and a positive integer F,*

$$F = \left\lceil \frac{(F^2+1)^t}{\binom{F^2+t}{t}} \right\rceil \quad \Longleftrightarrow \quad F = t!.$$

Proof. Put $H_t(F) = t! \prod_{i=1}^{t} (F^2+1)/(F^2+i)$. Each factor is at most one, so the ceiling relation gives $H_t(F) \leq F \leq t!$. If $F < t$, monotonicity in F^2 gives $H_t(F) \geq 2^t/(t+1) > t > F$, impossible for $t \geq 6$. For $F \geq t$, the product inequality gives

$$H_t(F) > t! \left(1 - \frac{t(t-1)}{2F^2}\right) > \frac{t!}{2}.$$

If $F < t^2$, then $t!/2 > t^2$ gives another contradiction. Thus $F \geq t^2$, and $H_t(F) > t!(1 - 1/(2F))$. If $t! \geq F+1$, the last lower bound is at least $(F+1)(1-1/(2F)) \geq F$, again impossible. Hence $F = t!$. Conversely at $F = t!$ the same estimates give $F - 1/2 < H_t(F) \leq F$, proving the ceiling relation. □

This factorial lemma is prior work in the repository and the shared-binomial parent; it is reproduced to make the dependency order transparent. Equations (12) and (14) now give $F = t!$.

5.1 All later digits and the smaller amplification

Every ordinary upper representative a satisfies $a \leq S_0$. Every requested lower index k is less than $R = K$. Thus all its required binomial coefficients are bounded by

$$a^k \leq (S_0+1)^k < (S_0+1)^K \leq q.$$

For the derived label polynomial, the retained literal positive-DAG majorant is $J \leq (S_0+1)^{\ell_0}$. At $t = 12$,

$$\ell_0(t+1) = 2^{22} \cdot 13 < 12! = 479,001,600.$$

The inequality persists under increasing t, so $\ell_0(t+1) < t! = F < R = K$. Hence its digits through place $t+1$ are also strictly below q.

The local evaluation bound is less than R^{E_0}. For $R \geq 12! > 2^{28}$,

$$R^{2^{23}} < 2^R.$$

One elementary verification takes $2^m \leq R < 2^{m+1}$, $m \geq 28$. The inequality $(m+1)2^{23} < 2^m$ holds at $m = 28$ and persists by induction. It follows that $R^{2^{23}} < 2^{(m+1)2^{23}} < 2^R$. Consequently amplification $K = R$ is sufficient; the former factor of two is unnecessary on this horizon. These estimates occur after the factorial has been recovered, while (13) occurs before it.

6 Modular numerators, exact powers, and CRT decoding

6.1 Recovering all exact powers

For a retained request $m = b^k$, introduce the nonnegative modulus $T_b = 2Ab - b^2 - 1$ and impose

$$X_k(A) + bY_k(A) = AY_k(A) + m + T_b h, \qquad h \geq 0.$$

The raw modulus row excludes $b = 0$. The mixed frame places $b, k, m < W$, and (7) makes T_b larger than both m and b^k. The Pell recurrence modulo T_b gives

$$X_k(A) + (b - A)Y_k(A) \equiv b^k \pmod{T_b}.$$

By Theorem 4.1, this is the actual requested k, not an unspecified congruent index. The strict interval then forces $m = b^k$. The positive expression on the left also supplies the required natural quotient in the constructive converse. Thus the exact powers can be decoded before the first binomial is interpreted.

6.2 One freely chosen root for every numerator

Let $b = q + 1$. The retained common modulus M is a product of the digit denominator powers and q, large enough that every required qd and Nd divides M. Its prime divisors divide q. Add

$$2A(q + 1) = (q + 1)^2 + 1 + Mh_0, \qquad h_0 \geq 0. \tag{15}$$

Then $s = q + 1 - A$ satisfies $s^2 \equiv A^2 - 1 \pmod{M}$. The identity $(A + s)^r = X_r(A) + sY_r(A)$ in the quadratic algebra yields

$$\mathcal{R}_r := X_r(A) + (q + 1 - A)Y_r(A) \equiv (q + 1)^r \pmod{M}. \tag{16}$$

Moreover $\mathcal{R}_r > 0$ for $r > 0$, because $X_r(A) > (A - 1)Y_r(A)$ and $q > 0$.

For the exact binomial at upper index $r = F^2 + t$ and denominator $d = q^t$, use

$$\mathcal{R}_r = r_0 + dB + qdh, \qquad d = r_0 + s_0 + 1, \qquad q = B + o + 1.$$

For a range request $N \mid \binom{r}{k}$, use

$$\mathcal{R}_r = r_0 + Ndh, \qquad d = q^k = r_0 + s_0 + 1.$$

The carry bounds of the preceding section and the binomial theorem identify these as the desired exact digit and divisibility conditions. The upper index of every such request is one of the newly included atoms in (5), so its Pell pair has already been certified by the same single anchor.

Completeness of (15) is not automatic from its notation. Since $\gcd(q + 1, M) = 1$, the gcd of $2(q + 1)$ and M is at most 2. If it is 2, $q + 1$ is odd and $(q + 1)^2 + 1$ is even. Thus the linear congruence for A is soluble. Its solution class contains arbitrarily large positive integers. After all semantic data, W, and L are fixed, choose such an $A \geq L$ large enough that $h_0 \geq 0$. All Pell and cap witnesses are then constructed at this chosen base. The positive residues are congruent to the least prescribed residues modulo their moduli, so the remaining quotients are nonnegative.

6.3 The reciprocal product and bounded source

Now $F = t!$ and $B = \binom{F^2+t}{t}$. For $1 \le i \le t$, define $d_i = 1 + F^2/i$. They are integers, pairwise coprime, their prime factors exceed t, and

$$\prod_{i=1}^{t} d_i = B.$$

The retained CRT source imposes, for its encoded index C and eight encoded columns A_j,

$$B \mid C + F^2 + 1, \qquad N \mid \binom{C}{t}, \qquad N \mid \binom{A_j}{b_j + 1},$$

$$N \mid L(C, \mathbf{A}), \qquad N \mid \binom{J(\mathbf{A})}{t+1}.$$

The one-way binomial range lemma states that $p^e \mid \binom{a}{k}$ implies $a \bmod p^e < k$. For $a \ge k$, its proof uses the usual floor-sum valuation formula: if $[a-k+1, a]$ contains no multiple of p^e, the valuation is at most $e-1$. The case $a < k$ is immediate. The false converse is never used.

Apply this at a prime factor of each d_i, with precision multiplied by K. It recovers the index $i-1$ and the bounded local witnesses. Since $2^K > R^{E_0}$, divisibility of the nonnegative local evaluation forces its actual value to be zero. The derived range yields the label bounds. This reconstructs every bounded source row and hence the finite refutation described in Section 3.

In the converse direction, use CRT at twice the valuation precision. For each prime p of d_i, the numerator of the relevant falling factorial has a factor divisible by $p^{2Kv_p(d_i)}$. Division by $k!$ loses at most $k-1 < K$ in valuation, leaving the required precision. Choose positive representatives and only then form their sum S_0 and the radix (11). This removes the old need to fit those representatives into an independently chosen aggregate bound.

7 Literal polynomial compilation and verification

Every formula above is translated into a finite quadratic graph using addition and multiplication nodes, norm rows, remainder equations, and the fixed local calculus polynomial. No factorial, exponentiation, binomial coefficient, Pell function, bounded quantifier, or divisibility symbol remains in the final list of residuals.

The principal source contains 777 semantic coordinates and 725 rows. An independently checked affine export restores 428 eliminated coordinates by polynomials known to be nonnegative on the natural domain. This gives 349 coordinates and 297 quadratic residuals. A further exact certificate eliminates 108 coordinates, leaving 241 coordinates and 189 residuals. Each elimination has a displayed restoration formula and a literal polynomial check. Preserving natural-valued restorations is essential: an arbitrary integer-valued substitution would not preserve the stipulated domain.

The compiler's support threshold is an implementation choice controlling which quadratic substitutions are retained. The selected threshold 32 gives the principal equation. Larger thresholds give the denser 239-variable companion. The algorithm is not claimed to find an optimal elimination order or a globally minimal graph.

All remaining residuals have degree at most two. At least one has a nonzero quadratic part, so the sum of their squares has exact degree four: over the reals a sum of squares of homogeneous quadratic forms cannot vanish identically unless every form is zero. The fully collected expansion independently confirms the degree and gives the exact term and height counts in Section 1.

The checks supplied with the packet cover the literal source rows, mixed frame atom set, anchor certificate, positive independent indices, pooled error congruence, ordinary polynomial expansion, and every deleted-variable restoration. Rebuilding the selected files in a fresh directory must give the same mathematical artifacts. Random numerical evaluations are used only as supplementary transcription checks where indicated; they are not the proof of an unbounded arithmetic claim or of the consistency endpoint.

8 A short ten-positive-variable companion

The direct quartic is the principal result. A separate improvement on the compact native branch uses a coordinate condition already forced at every zero. This branch retains the native/K215 calibration and its four large fixed coefficients, unlike the direct quartic.

Let (β, K_0, M_C, M_F) be the complete original compact44 tuple supplied in `positive/coefficients`. These are fixed integers, not unknowns. Their raw binary sizes are 590,069, 592,884, 590,069, and 590,070 bits respectively; $\beta = 2^{590069} - 1$. The entire tuple is retained, with M_F already shifted exactly once. Its payloads and finite reconstruction data are supplied together.

The ten unknowns are

$$J, F, \alpha, i, T, w, s, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

Define the following polynomial abbreviations:

$$\begin{aligned}
q &= \beta J + 1, & g &= q - 1, & X &= wq, & Y &= q^3 s,\\
\mu &= q^2 + 1, & k &= \eta + \zeta, & c &= kY + \eta,\\
A &= \mu Y(X+1) + 1, & \Delta &= A^2 - 1, & H &= 4A - 5,\\
V_1 &= X\mu^2 Y^2, & D &= V_1(V_1+1)k^2 + 1,\\
C &= q - F - \alpha, & B &= (K_0 + w)C - F,\\
G &= (q-1)(q-F) + \alpha, & R &= G(q^2-1) + (M_C + qM_F)J,\\
e &= k - R - 1, & E &= ic^2,\\
F_s &= 1 + \Delta D E^2, & Q &= \Delta^2 D E^2, & \delta &= Q - 1,\\
b &= c(T+\alpha) - R, & V &= F_s b - c, & D_* &= QV^2 - 1,\\
K_+ &= X(3c - 2X) + 2, & M &= g\mu^2 H, & a &= DF_s.
\end{aligned}$$

For the shortest version put

$$Z = g\mu^2\big[H(n + V - bB) + K_+\big] - H\big(\mu^2 B + ge\big). \tag{17}$$

The equation is the explicit polynomial

$$\boxed{\mathcal{N} = \big(\delta Z^2 - M^2 D_* - \delta M^2 a\big)^2 - 4\delta M^4 a D_* = 0.} \tag{18}$$

Every exponent in these definitions is a fixed nonnegative integer. There are no radicals or divisions in the equation. The fully paid division-free schedule takes **86 binary operations**: 51 multiplications, 18 additions, and 17 subtractions, including constant multiplications and squaring. Its exact total degree is **466**.

8.1 Quotient chart and exact cancellation

The pinned uncentered gap-fusion parent takes 89 operations and has degree 288 [4]. At every positive zero its proof yields $q^3 \mid Y$. Substitute $Y = q^3 s$ into that polynomial. Both its carrier numerator and denominator acquire a factor q^3, so its quartic norm acquires q^{12}. Canceling that explicit nonzero factor yields a polynomial of degree 466 with 87 operations. This version is in exact bijection with the parent's positive zero set: it retains $n + s$ in place of n in (17).

The shortest 86-operation version absorbs the integer term s into the positive padding coordinate. The exact universal polynomial identity is

$$N_{\mathrm{parent}}(Y = q^3 s,\ n_{\mathrm{old}} = n - s) = q^{12}\mathcal{N}. \tag{19}$$

The identity alone is not enough for soundness because $n - s$ could be nonpositive. The following additional argument handles that point.

8.2 Why absorbing the padding is sound

All pre-transport steps of the parent proof are independent of the sign of its padding variable. At a positive zero of (18) they recover positive integer roots τ, f, y with

$$D = \tau^2, \quad F_s = f^2, \quad QV^2 - (Q-1)y^2 = 1,$$

and give the early estimates

$$b > 8Q, \qquad \tau f < b/2, \qquad 0 < y - V < b/4, \qquad e/\mu^2 < b/4, \qquad K_+ > 0.$$

The fifth-index gap underlying $b > 8Q$ is prior work of the parent, not a new lemma here. The two-root norm gives, for signs ε, ν,

$$\frac{Z}{M} = \varepsilon\tau f + \nu y, \qquad \frac{Z}{M} = n + V + \frac{K_+}{H} - \frac{(1+gb)B}{g} - \frac{e}{\mu^2}.$$

Therefore

$$n = \varepsilon\tau f + (\nu y - V) - \frac{K_+}{H} + \frac{(1+gb)B}{g} + \frac{e}{\mu^2} < b + \frac{(1+gb)B}{g}.$$

If $B \le -1$, the right side is at most $-1/g < 0$, contradicting $n > 0$. Thus the same transport sign $B \ge 0$ is forced.

The subsequent rank, index, parity, projection, and source-reconstruction arguments use the integral carrier and this sign. The condition $q^3 \mid Y$ is now built into the coordinates; the remaining denominator is $g\mu^2 H$, strengthening the relevant H-divisibility statement. The whole proof recovers the same native source. Conversely a canonical parent witness has $Y = q^3 s$ with $s > 0$, and setting $n = n_{\mathrm{old}} + s$ is positive and satisfies (19). This is an existence-preserving reparameterization. We do not assert that its unshifted inverse sends every zero to a positive old padding coordinate.

8.3 Exact degree and the tradeoff

All ten unknowns have degree one, while the coefficient ports have degree zero. In the quotient chart the relevant degrees are

register	Y	A	Δ	D	E	F_s	Q	b	V	M	a	Z_0
degree	4	8	16	30	11	68	84	6	74	13	98	≤ 21,

where $Z = MV + Z_0$. Expand the norm after this substitution. Its unique highest-degree contribution is

$$-4\delta^2 M^4 V^2 a,$$

with degree $2 \cdot 84 + 4 \cdot 13 + 2 \cdot 74 + 98 = 466$. More explicitly its nonzero leading homogeneous form is

$$-2^{10}(\beta J)^{326} i^{10} w^{30} s^{64} (\eta + \zeta)^{34} (T + \alpha)^2.$$

The leading form of b is $(\eta + \zeta)Y_0(T + \alpha)$, not the old leading form $-R_0$; this change must be included when checking the new degree.

Adding the sharp product center ADE gives companion schedules with degree 450 and 89 operations (absorbed padding) or 90 operations (exact parent-positive bijection). The factor $\Gamma^2 - a = D^2E^2 - D$ lowers the leading contribution by 16. These are diagnostic controls: the 89-operation, degree-450 version is dominated by the already existing 89-operation, degree-288 equation. The 86-operation equation improves arithmetic length at the same ten-positive-variable count.

9 A ten-positive-variable equation of degree 246

A second native companion improves degree rather than operation count. It has ten positive unknowns, exact degree **246**, and a completely paid schedule of **121 operations** (73 multiplications, 25 additions, and 23 subtractions). Its parent sharp-center equation has degree 254 and 97 operations [4]. This is a degree/cost tradeoff. The following construction uses the same complete compact44 tuple as Section 8, but restores the independent coordinate Y: *the substitution $Y = q^3 s$ is not made here.*

9.1 The full polynomial and its two correction terms

Let $J, F, \alpha, i, T, x, Y, n, \eta, \zeta \in \mathbb{Z}_{>0}$. Define

$$\begin{aligned}
q &= \beta J + 1, & X &= q + x, & g &= q(q-1),\\
C_d &= q - F - \alpha, & B &= (qK_0 + X)C_d - qF, & &\\
\mu &= q^2 + 1, & t &= \mu Y, & L &= X + 1,\\
A &= tL + 1, & \Delta &= A^2 - 1, & H &= 4A - 5,\\
k &= \eta + \zeta, & c &= kY + \eta, & v &= Xt^2,\\
D &= 1 + v(v+1)k^2, & E &= ic^2, & &\\
F_s &= 1 + \Delta DE^2, & Q &= \Delta^2 DE^2, & \delta &= Q - 1.
\end{aligned}$$

The remaining unchanged kernel and carrier data are

$$\begin{aligned}
R &= [(q-1)(q-F) + \alpha](q^2-1) + (M_C + qM_F)J,\\
e &= k - R - 1, & b &= c(T+\alpha) - R, & V &= F_s b - c,\\
D_* &= QV^2 - 1, & P &= q^4(q-1), & M &= P\mu^2 H,\\
K_+ &= X(3c - 2X) + 2, & W &= 1 + gb. &&
\end{aligned}$$

The new polynomial abbreviations are

$$Q_1 = k^2\big[t^2 L X^2 (tL-1) + t(L^3 - 2L + 1) - L^2 + 2\big], \tag{20}$$

$$Q_3 = k^2(tX^2 - 3L + 6), \qquad \lambda = 8L^3, \tag{21}$$

$$\Gamma = E(8L^3 AD - 4Q_1 - Q_3), \qquad a = \lambda^2 DF_s. \tag{22}$$

Set

$$\begin{aligned} Z = {} & P\mu^2\big[H(n+V+\Gamma-bB)+K_+\big] \\ & + H\big[\mu^2(gY-q^3B-(q-1)X)-Pe\big]. \end{aligned} \tag{23}$$

The complete equation is

$$\boxed{\mathcal{N}_{246} = \left(\delta Z^2 - M^2 D_* - \delta M^2 a\right)^2 - 4\delta M^4 a D_* = 0.} \tag{24}$$

All abbreviations are integer polynomials in the ten listed unknowns. The corresponding literal graph and coefficient payloads are in `center246/`. No additional quantified coordinate is hidden in (24).

The construction comes from two exact polynomial divisions. With $\Omega = L^3 D$ and $X = L - 1$, expansion gives

$$\begin{aligned} &\Omega = AQ_1 + r_1 = A^3 Q_3 + r_3, \\ &r_1 = L^3 + k^2(L^2 - t - 2), \\ &r_3 = L^3 + k^2\big[L^2t^2(L^2+5L-3t-12) + Lt(8L-3t-16) + 3L - t - 6\big]. \end{aligned} \tag{25}$$

Thus $8\Omega A - 4Q_1 - Q_3$ retains the polynomial portions of the first three terms of $8\Omega\sqrt{A^2-1} = 8\Omega A - 4\Omega/A - \Omega/A^3 - \cdots$. The series is a discovery guide only: the construction and proof use the exact identities (25). For a shorter evaluation schedule, put $p = A - 1$, $u = tX^2$ and use

$$4Q_1 + Q_3 = k^2\big[u\{4p(p-1) + 4L + 9\} + 4(tX - L^2) - 3L + 14\big]. \tag{26}$$

9.2 Positive bounds, root recovery, and transport

Lemma 9.1. *On the whole positive input domain, $0 < \Gamma < Q$. At every positive zero of (24), the recovered Pell roots satisfy $\lambda\tau f + X/q^4 < 2Q < b/4$.*

Proof. Here $L \geq 33$ and $t \geq 1025$. Both Q_1, Q_3 are positive: the last part of Q_1/k^2 is at least $L^3 - L^2 - 2L + 3 > 0$, while $Q_3/k^2 \geq (L-2)(L-3) + 1 > 0$. The first term of Q_1/k^2 is at least $24t^3 > t + 2$, so $r_1 > -k^2(t+2) > -Q_1$ and (25) gives $\Omega > (A-1)Q_1 > Q_1$. Also $Q_3 < k^2 t X^2 < D \leq \Omega$. It follows that

$$0 < \Gamma < \lambda ADE, \qquad \lambda A = 8L^3(tL+1) \leq 16tL^4 < t^4L^4 < \Delta^2.$$

Since $E \geq 1$, this proves $\Gamma < Q$.

For completeness, root recovery does not require the desired transport sign. Since $F_s > c + 1$ and b is integral, $|V| > 1$ and $D_*/\delta > 0$. The positive radicands a and D_*/δ are unequal: equality would give $\delta\lambda^2 D F_s = QV^2 - 1$, impossible modulo $D > 1$. Writing $z = Z/M$, the norm therefore has $z \neq 0$ and gives the rational square root $(z^2 + a - D_*/\delta)/(2z)$ of a. As $\lambda > 0$ and $\gcd(D, F_s) = 1$, it follows that $D = \tau^2$ and $F_s = f^2$ for positive integers τ, f. With $I = \tau E$, the unchanged adjacent-Pell descent gives an integer $y > 0$ and

$$f^2 - \Delta I^2 = 1, \qquad Q = (\Delta I)^2, \qquad QV^2 - (Q-1)y^2 = 1, \qquad z = \varepsilon\lambda\tau f + \nu y.$$

Here $\varepsilon, \nu \in \{1, -1\}$. The parent's pre-transport folding, first-index, and fifth-index arguments now give $b > 8Q$, $K_+ > 0$, $0 < y - V < b/4$, and $e/\mu^2 < b/4$ [4]. None uses the new integral center or radical weight. Finally $f < AI$ implies $\lambda\tau f < \lambda ADE < Q$; also $X/q^4 < Q$. Their sum is less than $2Q < b/4$. □

The exact normalized carrier is

$$\frac{Z}{M} = n + V + \Gamma + \frac{K_+}{H} + \frac{Y}{q^3} - \frac{WB}{g} - \frac{X}{q^4} - \frac{e}{\mu^2}. \tag{27}$$

Combining it with the recovered signed roots and Lemma 9.1 gives

$$n < \frac{3b}{4} + \frac{WB}{g}.$$

If $B \le -1$, then $WB/g \le -b - 1/g$, contradicting $n > 0$. Thus $B \ge 0$ and $C_d > 0$. The remaining ordered rank, parity, coprimality, whole-product projection, and unscaled source reconstruction are those of the pinned gap-fusion proof. The center and weighted root are integral; the fractional terms in (27) and $W \equiv 1 \pmod g$ are unchanged. Hence the same source61/native/controller calibration recovers a finite ZFC refutation. This is the inherited native branch, not the direct C222 route of the principal quartic.

Conversely take the parent's fresh canonical kernel witnesses from a refutation. They supply integral $t_0 = B/g \ge 0$, $\rho = e/\mu^2 \ge 1$, $s_Y = Y/q^3 > 0$, $\kappa = K_+/H > 0$, $X/q^4 > 0$, and $u = y - V > 2Q$, with $s_Y, \kappa < c$. Define the remaining coordinate by

$$n = Wt_0 + \rho + \lambda\tau f - \Gamma + u - s_Y - \kappa + X/q^4. \tag{28}$$

It is an integer and is positive because $n > 2Q - Q - 2c > 0$; here $Q > c^4 > 2c$ follows from $E = ic^2$ and $c \ge 3$. The carrier becomes $\lambda\tau f + y$, so the norm vanishes. This proves existence equivalence by a fresh positive construction, without claiming a positive pointwise inverse for every parent zero.

9.3 Exact degree and a noncancelling coefficient

Write $C_* := \Gamma^2 - a$. Assign weights $1, 3, 1, 5$ to the independent polynomial variables L, t, k, E. Then $\deg A = 4$, $\deg D = 16$, $\deg \Omega = 19$, $\deg Q_1 = 15$, $\deg Q_3 = 7$, $\deg r_1 = 5$, and $\deg r_3 = 13$. An exact identity derived from (25) is

$$\begin{aligned} A^2(C_* + 64L^6D) = E^2\big[&64A^2\Omega r_1 + 16\Omega(r_3 - 2r_1) + 16r_1^2 \\ &+ 8A^2Q_1Q_3 + A^2Q_3^2\big]. \end{aligned} \tag{29}$$

The two degree-42 terms on the right have combined leading part $-112L^7t^7k^4E^2$. All others have smaller degree. Since $A_0 = Lt$, this proves

$$\deg C_* = 34, \qquad (C_*)_0 = -112L_0^5t_0^5k_0^4E_0^2. \tag{30}$$

The supplied certificate independently lists and checks all 52 coefficients of C_*.

In the actual ten inputs, the unchanged kernel has

$$\deg \delta = 42, \quad \deg V = 38, \quad \deg M = 13, \quad \deg a = 56, \quad \deg \Gamma = 28.$$

Write $Z = M(V + \Gamma) + Z_0$, where $\deg Z_0 \le 19$ and $(Z_0)_0 = -M_0b_0B_0$. Put

$$\Theta = \delta M^2C_* - M^2(V^2 - 1) + 2\delta M(V + \Gamma)Z_0 + \delta Z_0^2.$$

Direct expansion gives

$$\begin{aligned} \mathcal{N}_{246} = &4\delta^2M^4V^2C_* + 4\delta M^2V\Gamma\Theta + \Theta^2 \\ &- 4\delta M^4aV^2 + 4\delta M^4a. \end{aligned} \tag{31}$$

Here $\deg \Theta = 112$, so the five displayed terms have degrees at most $246, 246, 224, 226, 150$. The first two terms both contribute at degree 246; their possible cancellation must be checked.

Let $q_0 = \beta J$, $X_0 = q_0 + x$, $t_0 = q_0^2 Y$, $k_0 = \eta + \zeta$, $E_0 = ik_0^2 Y^2$, $b_0 = -q_0^3(q_0 - F)$, and $B_0 = (q_0 K_0 + X_0)(q_0 - F - \alpha) - q_0 F$. The complete leading form is

$$4\delta_0^2 M_0^4 V_0^2((C_*)_0 - 2\Gamma_0 b_0 B_0), \qquad \Gamma_0 = 8X_0^6 t_0^5 k_0^2 E_0.$$

Its first summand has degree ten in i and equals

$$-7 \cdot 2^{14} q_0^{116} X_0^{29} i^{10} Y^{57} k_0^{32} (q_0 - F)^2;$$

the second has degree nine in i. Consequently

$$\boxed{[J^{116} x^{29} i^{10} Y^{57} \eta^{32} F^2]\,\mathcal{N}_{246} = -7 \cdot 2^{14} \beta^{116} \neq 0.} \tag{32}$$

This proves exact degree 246 for the actual fixed tuple, rather than only for generic ports.

The companion checker passes 151 checks, including exact local polynomial identities, complete coefficient and source pins, circuit liveness, independent formula evaluations, and degree/leading-form transcription checks over three finite fields. Polynomial-part radical centering already appears elsewhere in ProveIt, for example the unrestricted-integer construction in `87-direct-centered-round12/norm/PROOF.md`. The new result is this concrete application and degree/cost point relative to the frozen snapshot. It inherits the parent's ordinary calibration; no new end-to-end proof-assistant certificate or global minimality claim is made.

10 What is new, what is inherited, and what remains open

The contribution of this round is a proved and fully materialized combination, together with a new use of a raw bootstrap inequality. The single anchor must include modular indices, the self-sized radix is justified by $V_0 \leq BF$ before factorial semantics, and the complete contracted quartic is counted and supplied. Merely placing two prior algorithms next to one another would not establish any of those points.

The factorial fixed-point lemma, common modular Pell residues, binomial range method, reciprocal CRT source, one-anchor error-square theorem, and natural-variable contraction each have explicit predecessors in the retained sources. The relation-combining and Pell tradition is classical. We make no claim to have invented those general techniques. The novelty audit is bounded to the inspected repository snapshot, incoming archives, and cited literature; it is not a worldwide priority search or a minimum-size proof.

The native companion establishes a different saving: a forced quotient condition can be built into the coordinates, a homogeneous clearing factor can be removed, and an integer term can be absorbed into a padding coordinate after re-proving its sign argument. The distinction between an exact positive-zero bijection and fresh positive completeness is essential in that last step.

There are still concrete opportunities. A better contraction order may reduce the direct quartic without changing the number theory. A smaller direct proof calculus may simplify its local source, provided its full refutation equivalence is proved. A common indexed bound tailored to the actual modular indices might reduce the remaining anchor overhead. None of these possibilities is included in the proved counts above.

The packet establishes an equivalence, not either side of it. In particular it neither finds a refutation nor proves that all tuples fail to solve the equation. A newly checked, end-to-end formal proof remains a separate task. The delivered finite objects and explicit proof boundaries are intended to make that task and further optimization concrete.

A Every residual of the principal quartic

The following is the complete definition of the polynomials Q_j in (1). Indices are zero-based. All z_i independently range over $\mathbb{N}$. Line breaks within one Q_j are typographical only. The appendix is generated directly from the authoritative equation file and checked against it by the release verifier.

Residual	Integer polynomial (continued across rows when necessary)
$Q_0 =$	$-z_2^2 + z_3$
$Q_1 =$	$1 - 2z_2 - z_2z_4 + z_4 + z_5 - z_6$
$Q_2 =$	$1 + z_4 - z_6 - z_7$
$Q_3 =$	$-13 - z_0 - z_0z_2 - z_0z_9 - 13z_2 - 13z_9$ $+z_{11}$
$Q_4 =$	$-1 + z_1 - z_2 - z_2z_8 - 2z_8 - z_8z_9$
$Q_5 =$	$-1 + z_1 - z_{10} - z_{10}z_{11}$
$Q_6 =$	$2 + z_3 - z_4z_{22} + z_{20} - 2z_{22}$
$Q_7 =$	$-6 - 5z_{12} - z_{12}^2 - 2z_{12}z_{13} - 5z_{13} - z_{13}^2$ $+z_{23}$
$Q_8 =$	$-72 - 68z_{13} - 16z_{13}^2 - 16z_{13}z_{23} - 34z_{23} - 4z_{23}^2$ $+z_{24}$
$Q_9 =$	$-30 - 11z_{12} - z_{12}^2 - 4z_{12}z_{24} - 22z_{24} - 4z_{24}^2$ $+z_{25}$
$Q_{10} =$	$-380 - 312z_{24} - 64z_{24}^2 - 32z_{24}z_{25} - 78z_{25} - 4z_{25}^2$ $+z_{26}$
$Q_{11} =$	$-6 - 5z_{13} - z_{13}^2 - 2z_{13}z_{14} - 5z_{14} - z_{14}^2$ $+z_{27}$
$Q_{12} =$	$-6 - 5z_{12} - z_{12}^2 - 2z_{12}z_{14} - 5z_{14} - z_{14}^2$ $+z_{28}$
$Q_{13} =$	$-414 - 22z_{12} - 2z_{12}^2 - 16z_{12}z_{14} - 8z_{12}z_{27} - 200z_{13}$ $-32z_{13}^2 - 64z_{13}z_{14} - 32z_{13}z_{23} - 32z_{13}z_{28} - 320z_{14} - 64z_{14}^2$ $-32z_{14}z_{23} - 32z_{14}z_{27} - 32z_{14}z_{28} - 100z_{23} - 8z_{23}^2 - 16z_{23}z_{28}$ $-52z_{27} - 8z_{27}^2 - 108z_{28} - 8z_{28}^2 + z_{29}$
$Q_{14} =$	$-1344 - 800z_{13} - 128z_{13}^2 - 256z_{13}z_{14} - 128z_{13}z_{23} - 128z_{13}z_{28}$ $-864z_{14} - 128z_{14}^2 - 128z_{14}z_{23} - 128z_{14}z_{28} - 400z_{23} - 32z_{23}^2$ $-64z_{23}z_{28} - 432z_{28} - 32z_{28}^2 - 2z_{29} - 2z_{29}^2 + z_{30}$
$Q_{15} =$	$-247 - 168z_{12} - 32z_{12}^2 - 64z_{12}z_{13} - 32z_{12}z_{27} - 32z_{12}z_{28}$ $-188z_{13} - 32z_{13}^2 - 32z_{13}z_{27} - 32z_{13}z_{28} - 2z_{23} - 92z_{27}$ $-8z_{27}^2 - 16z_{27}z_{28} - 84z_{28} - 8z_{28}^2 + z_{31}$
$Q_{16} =$	$-960 - 672z_{12} - 128z_{12}^2 - 256z_{12}z_{13} - 128z_{12}z_{27} - 128z_{12}z_{28}$ $-736z_{13} - 128z_{13}^2 - 128z_{13}z_{27} - 128z_{13}z_{28} - 368z_{27} - 32z_{27}^2$ $-64z_{27}z_{28} - 336z_{28} - 32z_{28}^2 - 2z_{31} - 2z_{31}^2 + z_{32}$
$Q_{17} =$	$-276 - 30z_{12} - 2z_{12}^2 - 16z_{12}z_{14} - 8z_{12}z_{27} - 30z_{13}$ $-2z_{13}^2 - 16z_{13}z_{14} - 8z_{13}z_{28} - 272z_{14} - 64z_{14}^2 - 32z_{14}z_{27}$ $-32z_{14}z_{28} - 68z_{27} - 8z_{27}^2 - 68z_{28} - 8z_{28}^2 + z_{33}$
$Q_{18} =$	$-552 - 120z_{13} - 8z_{13}^2 - 64z_{13}z_{14} - 32z_{13}z_{28} - 544z_{14}$ $-128z_{14}^2 - 128z_{14}z_{28} - 272z_{28} - 32z_{28}^2 - 2z_{33} - 2z_{33}^2$ $+z_{34}$
$Q_{19} =$	$-247 - 4z_{12} - 352z_{13} - 128z_{13}^2 - 64z_{13}z_{23} - 64z_{13}z_{27}$ $-92z_{23} - 8z_{23}^2 - 16z_{23}z_{27} - 84z_{27} - 8z_{27}^2 - 2z_{28}$ $+z_{35}$
$Q_{20} =$	$-960 - 1408z_{13} - 512z_{13}^2 - 256z_{13}z_{23} - 256z_{13}z_{27} - 368z_{23}$

Residual	Integer polynomial (continued across rows when necessary)
	$-32z_{23}^2 - 64z_{23}z_{27} - 336z_{27} - 32z_{27}^2 - 2z_{35} - 2z_{35}^2$
	$+z_{36}$
$Q_{21} =$	$-56 - 60z_{13} - 16z_{13}^2 - 16z_{13}z_{23} - 30z_{23} - 4z_{23}^2$
	$+z_{37}$
$Q_{22} =$	$-30 - 11z_{12} - z_{12}^2 - 4z_{12}z_{37} - 22z_{37} - 4z_{37}^2$
	$+z_{38}$
$Q_{23} =$	$-380 - 312z_{37} - 64z_{37}^2 - 32z_{37}z_{38} - 78z_{38} - 4z_{38}^2$
	$+z_{39}$
$Q_{24} =$	$-30 - 11z_{13} - z_{13}^2 - 4z_{13}z_{39} - 22z_{39} - 4z_{39}^2$
	$+z_{40}$
$Q_{25} =$	$-349 - 420z_{13} - 128z_{13}^2 - 128z_{13}z_{23} - 208z_{23} - 32z_{23}^2$
	$-2z_{27} + z_{41}$
$Q_{26} =$	$-1368 - 1664z_{13} - 512z_{13}^2 - 512z_{13}z_{23} - 832z_{23} - 128z_{23}^2$
	$-2z_{41} - 2z_{41}^2 + z_{42}$
$Q_{27} =$	$-56 - 60z_{12} - 16z_{12}^2 - 16z_{12}z_{28} - 8z_{12}z_{42} - 30z_{28}$
	$-4z_{28}^2 - 4z_{28}z_{42} - 15z_{42} - z_{42}^2 + z_{43}$
$Q_{28} =$	$-6 - 5z_{14} - z_{14}^2 - 2z_{14}z_{15} - 5z_{15} - z_{15}^2$
	$+z_{44}$
$Q_{29} =$	$-224 - 26z_{12} - 2z_{12}^2 - 16z_{12}z_{15} - 8z_{12}z_{44} - 288z_{14}$
	$-128z_{14}^2 - 64z_{14}z_{27} - 64z_{14}z_{28} - 88z_{15} - 32z_{15}^2 - 32z_{15}z_{44}$
	$-76z_{27} - 8z_{27}^2 - 16z_{27}z_{28} - 68z_{28} - 8z_{28}^2 - 44z_{44}$
	$-8z_{44}^2 + z_{45}$
$Q_{30} =$	$-640 - 1152z_{14} - 512z_{14}^2 - 256z_{14}z_{27} - 256z_{14}z_{28} - 304z_{27}$
	$-32z_{27}^2 - 64z_{27}z_{28} - 272z_{28} - 32z_{28}^2 - 2z_{45} - 2z_{45}^2$
	$+z_{46}$
$Q_{31} =$	$-241 - 168z_{12} - 32z_{12}^2 - 64z_{12}z_{13} - 32z_{12}z_{27} - 32z_{12}z_{28}$
	$-184z_{13} - 32z_{13}^2 - 32z_{13}z_{27} - 32z_{13}z_{28} - z_{14} - 92z_{27}$
	$-8z_{27}^2 - 16z_{27}z_{28} - 84z_{28} - 8z_{28}^2 + z_{47}$
$Q_{32} =$	$-42 - 52z_{14} - 16z_{14}^2 - 16z_{14}z_{27} - 26z_{27} - 4z_{27}^2$
	$+z_{48}$
$Q_{33} =$	$-30 - 11z_{14} - z_{14}^2 - 4z_{14}z_{48} - 22z_{48} - 4z_{48}^2$
	$+z_{49}$
$Q_{34} =$	$-306 - 35z_{13} - z_{13}^2 - 16z_{13}z_{48} - 4z_{13}z_{49} - 280z_{48}$
	$-64z_{48}^2 - 32z_{48}z_{49} - 70z_{49} - 4z_{49}^2 + z_{50}$
$Q_{35} =$	$-4556 - 4320z_{48} - 1024z_{48}^2 - 512z_{48}z_{49} - 128z_{48}z_{50} - 1080z_{49}$
	$-64z_{49}^2 - 32z_{49}z_{50} - 270z_{50} - 4z_{50}^2 + z_{51}$
$Q_{36} =$	$-30 - 11z_{14} - z_{14}^2 - 4z_{14}z_{51} - 22z_{51} - 4z_{51}^2$
	$+z_{52}$
$Q_{37} =$	$-306 - 35z_{13} - z_{13}^2 - 16z_{13}z_{51} - 4z_{13}z_{52} - 280z_{51}$
	$-64z_{51}^2 - 32z_{51}z_{52} - 70z_{52} - 4z_{52}^2 + z_{53}$
$Q_{38} =$	$-1028 - 672z_{12} - 128z_{12}^2 - 256z_{12}z_{13} - 128z_{12}z_{27} - 128z_{12}z_{28}$
	$-736z_{13} - 128z_{13}^2 - 128z_{13}z_{27} - 128z_{13}z_{28} - 368z_{27} - 32z_{27}^2$
	$-64z_{27}z_{28} - 336z_{28} - 32z_{28}^2 - 2z_{47} - 2z_{47}^2 - 32z_{51}$
	$-8z_{52} - 2z_{53} + z_{54}$
$Q_{39} =$	$-264 - 128z_{51} - 32z_{52} - 8z_{53} - 2z_{54} - 2z_{54}^2$
	$+z_{55}$
$Q_{40} =$	$-2 - 3z_{12} - z_{12}^2 - 2z_{12}z_{55} - 3z_{55} - z_{55}^2$
	$+z_{56}$
$Q_{41} =$	$-12 - 28z_{55} - 16z_{55}^2 - 16z_{55}z_{56} - 14z_{56} - 4z_{56}^2$

Residual	Integer polynomial (continued across rows when necessary)
	$+z_{57}$
$Q_{42} =$	$-30 - 11z_{14} - z_{14}^2 - 4z_{14}z_{57} - 22z_{57} - 4z_{57}^2$ $+z_{58}$
$Q_{43} =$	$-306 - 35z_{12} - z_{12}^2 - 16z_{12}z_{57} - 4z_{12}z_{58} - 280z_{57}$ $-64z_{57}^2 - 32z_{57}z_{58} - 70z_{58} - 4z_{58}^2 + z_{59}$
$Q_{44} =$	$-4556 - 4320z_{57} - 1024z_{57}^2 - 512z_{57}z_{58} - 128z_{57}z_{59} - 1080z_{58}$ $-64z_{58}^2 - 32z_{58}z_{59} - 270z_{59} - 4z_{59}^2 + z_{60}$
$Q_{45} =$	$-90 - 76z_{14} - 16z_{14}^2 - 16z_{14}z_{27} - 16z_{14}z_{60} - 38z_{27}$ $-4z_{27}^2 - 8z_{27}z_{60} - 38z_{60} - 4z_{60}^2 + z_{61}$
$Q_{46} =$	$-4556 - 4320z_{51} - 1024z_{51}^2 - 512z_{51}z_{52} - 128z_{51}z_{53} - 1080z_{52}$ $-64z_{52}^2 - 32z_{52}z_{53} - 270z_{53} - 4z_{53}^2 + z_{62}$
$Q_{47} =$	$-682 - 70z_{13} - 2z_{13}^2 - 32z_{13}z_{60} - 8z_{13}z_{61} - 592z_{60}$ $-128z_{60}^2 - 64z_{60}z_{61} - 148z_{61} - 8z_{61}^2 - 2z_{62} + z_{63}$
$Q_{48} =$	$-z_{63} - z_{63}^2 + z_{64}$
$Q_{49} =$	$-42 - 52z_{13} - 16z_{13}^2 - 8z_{13}z_{14} - 16z_{13}z_{23} - 13z_{14}$ $-z_{14}^2 - 4z_{14}z_{23} - 26z_{23} - 4z_{23}^2 + z_{65}$
$Q_{50} =$	$-2278 - 2264z_{13} - 800z_{13}^2 - 800z_{13}z_{23} - 160z_{13}z_{65} - 70z_{14}$ $-2z_{14}^2 - 32z_{14}z_{62} - 8z_{14}z_{64} - 1132z_{23} - 200z_{23}^2 - 80z_{23}z_{65}$ $-592z_{62} - 128z_{62}^2 - 64z_{62}z_{64} - 148z_{64} - 8z_{64}^2 - 228z_{65}$ $-8z_{65}^2 + z_{66}$
$Q_{51} =$	$-2712 - 280z_{14} - 8z_{14}^2 - 128z_{14}z_{62} - 32z_{14}z_{64} - 2368z_{62}$ $-512z_{62}^2 - 256z_{62}z_{64} - 592z_{64} - 32z_{64}^2 - 2z_{66} - 2z_{66}^2$ $+z_{67}$
$Q_{52} =$	$-2 - 3z_{67} - z_{67}^2 + z_{68}$
$Q_{53} =$	$-z_2 - z_2z_{17} + z_{69}$
$Q_{54} =$	$-z_{16}^2 + 2z_{16}z_{17} - z_{17}^2 + z_{70}$
$Q_{55} =$	$-2z_{26}z_{69} - 4z_{69} + z_{71}$
$Q_{56} =$	$-z_{16}^2 + 2z_{16}z_{17} + 2z_{16}z_{71} - z_{17}^2 - 2z_{17}z_{71} - z_{71}^2$ $+z_{72}$
$Q_{57} =$	$-z_{30}z_{69} + z_{73}$
$Q_{58} =$	$-z_{16}^2 + 2z_{16}z_{17} + 2z_{16}z_{73} - z_{17}^2 - 2z_{17}z_{73} - z_{73}^2$ $+z_{74}$
$Q_{59} =$	$-z_{32}z_{69} + z_{75}$
$Q_{60} =$	$-z_{16}^2 + 2z_{16}z_{17} + 2z_{16}z_{75} - z_{17}^2 - 2z_{17}z_{75} - z_{75}^2$ $+z_{76}$
$Q_{61} =$	$-z_{34}z_{69} + z_{77}$
$Q_{62} =$	$-z_{16}^2 + 2z_{16}z_{17} + 2z_{16}z_{77} - z_{17}^2 - 2z_{17}z_{77} - z_{77}^2$ $+z_{78}$
$Q_{63} =$	$-z_{36}z_{69} + z_{79}$
$Q_{64} =$	$-z_{16}^2 + 2z_{16}z_{17} + 2z_{16}z_{79} - z_{17}^2 - 2z_{17}z_{79} - z_{79}^2$ $+z_{80}$
$Q_{65} =$	$-4z_{13}z_{69} - 2z_{40}z_{69} - 4z_{69} + z_{81}$
$Q_{66} =$	$-z_{16}^2 + 2z_{16}z_{17} + 2z_{16}z_{81} - z_{17}^2 - 2z_{17}z_{81} - z_{81}^2$ $+z_{82}$
$Q_{67} =$	$-4z_{42}z_{69} - 2z_{43}z_{69} + z_{83}$
$Q_{68} =$	$-z_{16}^2 + 2z_{16}z_{17} + 2z_{16}z_{83} - z_{17}^2 - 2z_{17}z_{83} - z_{83}^2$ $+z_{84}$

Residual	Integer polynomial (continued across rows when necessary)
$Q_{69} =$	$-z_{46}z_{69} + z_{85}$
$Q_{70} =$	$-z_{16}^2 + 2z_{16}z_{17} + 2z_{16}z_{85} - z_{17}^2 - 2z_{17}z_{85} - z_{85}^2$ $+z_{86}$
$Q_{71} =$	$-z_{12}^2 + 2z_{12}z_{14} - z_{14}^2 + z_{87}$
$Q_{72} =$	$-z_{13}^2 + 2z_{13}z_{14} - z_{14}^2 + z_{88}$
$Q_{73} =$	$-z_{87}z_{88} + z_{89}$
$Q_{74} =$	$-4 - 4z_{15} - z_{15}^2 + z_{90}$
$Q_{75} =$	$-z_{89}^2 + 2z_{89}z_{90} - z_{90}^2 + z_{91}$
$Q_{76} =$	$-2z_{68}z_{69} - 4z_{69} + z_{92}$
$Q_{77} =$	$-z_{16}^2 + 2z_{16}z_{17} + 2z_{16}z_{92} - z_{17}^2 - 2z_{17}z_{92} - z_{92}^2$ $+z_{93}$
$Q_{78} =$	$-z_2 - z_2z_{13} + z_{94}$
$Q_{79} =$	$-1 - z_{17} - z_{17}z_{94} - z_{94} + z_{95}$
$Q_{80} =$	$-z_2 - z_2z_{12} + z_{96}$
$Q_{81} =$	$-4z_2 - 4z_2z_{13} - 2z_2z_{23} + z_{97}$
$Q_{82} =$	$-1 - z_{16} - z_{16}z_{96} - z_{96} + z_{98}$
$Q_{83} =$	$-z_{97}z_{98} - z_{98} + z_{99}$
$Q_{84} =$	$-z_{95}^2 + 2z_{95}z_{99} - z_{99}^2 + z_{100}$
$Q_{85} =$	$-6z_2 - 4z_2z_{13} - 2z_2z_{23} + z_{101}$
$Q_{86} =$	$-1 - z_{17} - z_{17}z_{101} - z_{101} + z_{102}$
$Q_{87} =$	$-1 - z_{16} - z_{16}z_{94} - z_{94} + z_{103}$
$Q_{88} =$	$-z_{102}^2 + 2z_{102}z_{103} - z_{103}^2 + z_{104}$
$Q_{89} =$	$-z_{70}z_{72} + z_{105}$
$Q_{90} =$	$-z_{74}z_{105} + z_{106}$
$Q_{91} =$	$-z_{76}z_{106} + z_{107}$
$Q_{92} =$	$-z_{78}z_{107} + z_{108}$
$Q_{93} =$	$-z_{80}z_{108} + z_{109}$
$Q_{94} =$	$-z_{82}z_{109} + z_{110}$
$Q_{95} =$	$-z_{84}z_{110} + z_{111}$
$Q_{96} =$	$-z_{86}z_{111} + z_{112}$
$Q_{97} =$	$-z_{91}z_{112} - z_{93}z_{112} + z_{113}$
$Q_{98} =$	$-z_{100}z_{113} + z_{114}$
$Q_{99} =$	$-2 - 2z_2 - z_2z_{20} - 2z_9 - z_9z_{20} - z_{20}$ $+z_{115}$
$Q_{100} =$	$-1 - z_{18} - z_{18}z_{115} - z_{115} + z_{116}$
$Q_{101} =$	$-2 - z_2 - z_2z_{19} - z_9 - z_9z_{19} - 2z_{19}$ $-z_{19}z_{115} - z_{115} + z_{117}$
$Q_{102} =$	$2 - 4z_1 + 2z_1^2 - 2z_1z_{16} - 2z_1z_{17} - 2z_1z_{116}$ $-2z_1z_{117} + 2z_{16} + z_{16}^2 + 2z_{16}z_{116} + 2z_{17} + z_{17}^2$ $+2z_{17}z_{117} - z_{21}z_{118} + z_{104}z_{114} + 2z_{116} + z_{116}^2 + 2z_{117}$ $+z_{117}^2$
$Q_{103} =$	$3 + z_4 - z_{21}z_{119} + z_{173}$
$Q_{104} =$	$-2 - z_4 - z_4z_{171} - z_4z_{172} - 3z_{171} - z_{171}z_{173}$

Residual	Integer polynomial (continued across rows when necessary)
	$-3z_{172} - z_{172}z_{173} - z_{173} + z_{178} + z_{179}$
$Q_{105} =$	$-3 - z_4 - z_4z_{178} - z_4z_{179} + z_{120} - z_{173}$ $-z_{173}z_{178} - z_{173}z_{179} - 3z_{178} - 3z_{179}$
$Q_{106} =$	$-z_{120} - z_{120}z_{203} - z_{120}z_{204} + z_{121}$
$Q_{107} =$	$-z_{121} - z_{121}z_{216} - z_{121}z_{217} + z_{122}$
$Q_{108} =$	$-1 - z_{125} - z_{125}z_{216} - z_{125}z_{217} - z_{125}z_{240} + z_{126}$ $-z_{216} - z_{217} - z_{240}$
$Q_{109} =$	$-2z_{126} - z_{126}z_{216} - z_{126}z_{217} - z_{126}z_{240} + z_{127}$
$Q_{110} =$	$-1 + z_{123}^2 + z_{126}^2 - z_{127}^2$
$Q_{111} =$	$-17 - 8z_4 - z_4^2 + 2z_4z_{123} + 2z_4z_{124} - 2z_4z_{173}$ $-z_{122}z_{128} + 8z_{123} + 2z_{123}z_{173} + 8z_{124} + 2z_{124}z_{173} - 8z_{173}$ $-z_{173}^2$
$Q_{112} =$	$-8 - 8z_{134} - 4z_{134}z_{145} - 4z_{134}z_{150} - 4z_{134}z_{160} - 4z_{134}z_{164}$ $-4z_{134}z_{168} - 4z_{134}z_{175} - 4z_{134}z_{182} - 4z_{134}z_{188} - 4z_{134}z_{194} - 4z_{134}z_{200}$ $-4z_{134}z_{207} - 4z_{134}z_{213} - 4z_{134}z_{216} - 4z_{134}z_{217} - 4z_{134}z_{220} - 4z_{134}z_{226}$ $-4z_{134}z_{233} - 4z_{134}z_{238} - 4z_{134}z_{240} + z_{135} - 4z_{145} - 4z_{150}$ $-4z_{160} - 4z_{164} - 4z_{168} - 4z_{175} - 4z_{182} - 4z_{188}$ $-4z_{194} - 4z_{200} - 4z_{207} - 4z_{213} - 4z_{216} - 4z_{217}$ $-4z_{220} - 4z_{226} - 4z_{233} - 4z_{238} - 4z_{240}$
$Q_{113} =$	$-2z_{123} - z_{123}z_{145} - z_{123}z_{150} - z_{123}z_{160} - z_{123}z_{164} - z_{123}z_{168}$ $-z_{123}z_{175} - z_{123}z_{182} - z_{123}z_{188} - z_{123}z_{194} - z_{123}z_{200} - z_{123}z_{207}$ $-z_{123}z_{213} - z_{123}z_{216} - z_{123}z_{217} - z_{123}z_{220} - z_{123}z_{226} - z_{123}z_{233}$ $-z_{123}z_{238} - z_{123}z_{240} - 2z_{124} - z_{124}z_{145} - z_{124}z_{150} - z_{124}z_{160}$ $-z_{124}z_{164} - z_{124}z_{168} - z_{124}z_{175} - z_{124}z_{182} - z_{124}z_{188} - z_{124}z_{194}$ $-z_{124}z_{200} - z_{124}z_{207} - z_{124}z_{213} - z_{124}z_{216} - z_{124}z_{217} - z_{124}z_{220}$ $-z_{124}z_{226} - z_{124}z_{233} - z_{124}z_{238} - z_{124}z_{240} + z_{136}$
$Q_{114} =$	$3 + z_{129}^2 - z_{136}^2 + 4z_{145} + z_{145}^2 + 2z_{145}z_{150}$ $+2z_{145}z_{160} + 2z_{145}z_{164} + 2z_{145}z_{168} + 2z_{145}z_{175} + 2z_{145}z_{182} + 2z_{145}z_{188}$ $+2z_{145}z_{194} + 2z_{145}z_{200} + 2z_{145}z_{207} + 2z_{145}z_{213} + 2z_{145}z_{216} + 2z_{145}z_{217}$ $+2z_{145}z_{220} + 2z_{145}z_{226} + 2z_{145}z_{233} + 2z_{145}z_{238} + 2z_{145}z_{240} + 4z_{150}$ $+z_{150}^2 + 2z_{150}z_{160} + 2z_{150}z_{164} + 2z_{150}z_{168} + 2z_{150}z_{175} + 2z_{150}z_{182}$ $+2z_{150}z_{188} + 2z_{150}z_{194} + 2z_{150}z_{200} + 2z_{150}z_{207} + 2z_{150}z_{213} + 2z_{150}z_{216}$ $+2z_{150}z_{217} + 2z_{150}z_{220} + 2z_{150}z_{226} + 2z_{150}z_{233} + 2z_{150}z_{238} + 2z_{150}z_{240}$ $+4z_{160} + z_{160}^2 + 2z_{160}z_{164} + 2z_{160}z_{168} + 2z_{160}z_{175} + 2z_{160}z_{182}$ $+2z_{160}z_{188} + 2z_{160}z_{194} + 2z_{160}z_{200} + 2z_{160}z_{207} + 2z_{160}z_{213} + 2z_{160}z_{216}$ $+2z_{160}z_{217} + 2z_{160}z_{220} + 2z_{160}z_{226} + 2z_{160}z_{233} + 2z_{160}z_{238} + 2z_{160}z_{240}$ $+4z_{164} + z_{164}^2 + 2z_{164}z_{168} + 2z_{164}z_{175} + 2z_{164}z_{182} + 2z_{164}z_{188}$ $+2z_{164}z_{194} + 2z_{164}z_{200} + 2z_{164}z_{207} + 2z_{164}z_{213} + 2z_{164}z_{216} + 2z_{164}z_{217}$ $+2z_{164}z_{220} + 2z_{164}z_{226} + 2z_{164}z_{233} + 2z_{164}z_{238} + 2z_{164}z_{240} + 4z_{168}$ $+z_{168}^2 + 2z_{168}z_{175} + 2z_{168}z_{182} + 2z_{168}z_{188} + 2z_{168}z_{194} + 2z_{168}z_{200}$ $+2z_{168}z_{207} + 2z_{168}z_{213} + 2z_{168}z_{216} + 2z_{168}z_{217} + 2z_{168}z_{220} + 2z_{168}z_{226}$ $+2z_{168}z_{233} + 2z_{168}z_{238} + 2z_{168}z_{240} + 4z_{175} + z_{175}^2 + 2z_{175}z_{182}$ $+2z_{175}z_{188} + 2z_{175}z_{194} + 2z_{175}z_{200} + 2z_{175}z_{207} + 2z_{175}z_{213} + 2z_{175}z_{216}$ $+2z_{175}z_{217} + 2z_{175}z_{220} + 2z_{175}z_{226} + 2z_{175}z_{233} + 2z_{175}z_{238} + 2z_{175}z_{240}$ $+4z_{182} + z_{182}^2 + 2z_{182}z_{188} + 2z_{182}z_{194} + 2z_{182}z_{200} + 2z_{182}z_{207}$ $+2z_{182}z_{213} + 2z_{182}z_{216} + 2z_{182}z_{217} + 2z_{182}z_{220} + 2z_{182}z_{226} + 2z_{182}z_{233}$ $+2z_{182}z_{238} + 2z_{182}z_{240} + 4z_{188} + z_{188}^2 + 2z_{188}z_{194} + 2z_{188}z_{200}$ $+2z_{188}z_{207} + 2z_{188}z_{213} + 2z_{188}z_{216} + 2z_{188}z_{217} + 2z_{188}z_{220} + 2z_{188}z_{226}$ $+2z_{188}z_{233} + 2z_{188}z_{238} + 2z_{188}z_{240} + 4z_{194} + z_{194}^2 + 2z_{194}z_{200}$ $+2z_{194}z_{207} + 2z_{194}z_{213} + 2z_{194}z_{216} + 2z_{194}z_{217} + 2z_{194}z_{220} + 2z_{194}z_{226}$ $+2z_{194}z_{233} + 2z_{194}z_{238} + 2z_{194}z_{240} + 4z_{200} + z_{200}^2 + 2z_{200}z_{207}$ $+2z_{200}z_{213} + 2z_{200}z_{216} + 2z_{200}z_{217} + 2z_{200}z_{220} + 2z_{200}z_{226} + 2z_{200}z_{233}$ $+2z_{200}z_{238} + 2z_{200}z_{240} + 4z_{207} + z_{207}^2 + 2z_{207}z_{213} + 2z_{207}z_{216}$ $+2z_{207}z_{217} + 2z_{207}z_{220} + 2z_{207}z_{226} + 2z_{207}z_{233} + 2z_{207}z_{238} + 2z_{207}z_{240}$

Residual	Integer polynomial (continued across rows when necessary)
	$+4z_{213} + z_{213}^2 + 2z_{213}z_{216} + 2z_{213}z_{217} + 2z_{213}z_{220} + 2z_{213}z_{226}$
	$+2z_{213}z_{233} + 2z_{213}z_{238} + 2z_{213}z_{240} + 4z_{216} + z_{216}^2 + 2z_{216}z_{217}$
	$+2z_{216}z_{220} + 2z_{216}z_{226} + 2z_{216}z_{233} + 2z_{216}z_{238} + 2z_{216}z_{240} + 4z_{217}$
	$+z_{217}^2 + 2z_{217}z_{220} + 2z_{217}z_{226} + 2z_{217}z_{233} + 2z_{217}z_{238} + 2z_{217}z_{240}$
	$+4z_{220} + z_{220}^2 + 2z_{220}z_{226} + 2z_{220}z_{233} + 2z_{220}z_{238} + 2z_{220}z_{240}$
	$+4z_{226} + z_{226}^2 + 2z_{226}z_{233} + 2z_{226}z_{238} + 2z_{226}z_{240} + 4z_{233}$
	$+z_{233}^2 + 2z_{233}z_{238} + 2z_{233}z_{240} + 4z_{238} + z_{238}^2 + 2z_{238}z_{240}$
	$+4z_{240} + z_{240}^2$
$Q_{115} =$	$-z_{123}z_{131} - z_{124}z_{131} + z_{137}$
$Q_{116} =$	$-1 + z_{130}^2 + z_{131}^2 - z_{137}^2$
$Q_{117} =$	$-z_{133} - z_{133}z_{135} + z_{138}$
$Q_{118} =$	$-1 + z_{132}^2 + z_{133}^2 - z_{138}^2$
$Q_{119} =$	$-4 + z_{140} - 4z_{145} - z_{145}^2 - 2z_{145}z_{150} - 2z_{145}z_{160}$
	$-2z_{145}z_{164} - 2z_{145}z_{168} - 2z_{145}z_{175} - 2z_{145}z_{182} - 2z_{145}z_{188} - 2z_{145}z_{194}$
	$-2z_{145}z_{200} - 2z_{145}z_{207} - 2z_{145}z_{213} - 2z_{145}z_{216} - 2z_{145}z_{217} - 2z_{145}z_{220}$
	$-2z_{145}z_{226} - 2z_{145}z_{233} - 2z_{145}z_{238} - 2z_{145}z_{240} - 4z_{150} - z_{150}^2$
	$-2z_{150}z_{160} - 2z_{150}z_{164} - 2z_{150}z_{168} - 2z_{150}z_{175} - 2z_{150}z_{182} - 2z_{150}z_{188}$
	$-2z_{150}z_{194} - 2z_{150}z_{200} - 2z_{150}z_{207} - 2z_{150}z_{213} - 2z_{150}z_{216} - 2z_{150}z_{217}$
	$-2z_{150}z_{220} - 2z_{150}z_{226} - 2z_{150}z_{233} - 2z_{150}z_{238} - 2z_{150}z_{240} - 4z_{160}$
	$-z_{160}^2 - 2z_{160}z_{164} - 2z_{160}z_{168} - 2z_{160}z_{175} - 2z_{160}z_{182} - 2z_{160}z_{188}$
	$-2z_{160}z_{194} - 2z_{160}z_{200} - 2z_{160}z_{207} - 2z_{160}z_{213} - 2z_{160}z_{216} - 2z_{160}z_{217}$
	$-2z_{160}z_{220} - 2z_{160}z_{226} - 2z_{160}z_{233} - 2z_{160}z_{238} - 2z_{160}z_{240} - 4z_{164}$
	$-z_{164}^2 - 2z_{164}z_{168} - 2z_{164}z_{175} - 2z_{164}z_{182} - 2z_{164}z_{188} - 2z_{164}z_{194}$
	$-2z_{164}z_{200} - 2z_{164}z_{207} - 2z_{164}z_{213} - 2z_{164}z_{216} - 2z_{164}z_{217} - 2z_{164}z_{220}$
	$-2z_{164}z_{226} - 2z_{164}z_{233} - 2z_{164}z_{238} - 2z_{164}z_{240} - 4z_{168} - z_{168}^2$
	$-2z_{168}z_{175} - 2z_{168}z_{182} - 2z_{168}z_{188} - 2z_{168}z_{194} - 2z_{168}z_{200} - 2z_{168}z_{207}$
	$-2z_{168}z_{213} - 2z_{168}z_{216} - 2z_{168}z_{217} - 2z_{168}z_{220} - 2z_{168}z_{226} - 2z_{168}z_{233}$
	$-2z_{168}z_{238} - 2z_{168}z_{240} - 4z_{175} - z_{175}^2 - 2z_{175}z_{182} - 2z_{175}z_{188}$
	$-2z_{175}z_{194} - 2z_{175}z_{200} - 2z_{175}z_{207} - 2z_{175}z_{213} - 2z_{175}z_{216} - 2z_{175}z_{217}$
	$-2z_{175}z_{220} - 2z_{175}z_{226} - 2z_{175}z_{233} - 2z_{175}z_{238} - 2z_{175}z_{240} - 4z_{182}$
	$-z_{182}^2 - 2z_{182}z_{188} - 2z_{182}z_{194} - 2z_{182}z_{200} - 2z_{182}z_{207} - 2z_{182}z_{213}$
	$-2z_{182}z_{216} - 2z_{182}z_{217} - 2z_{182}z_{220} - 2z_{182}z_{226} - 2z_{182}z_{233} - 2z_{182}z_{238}$
	$-2z_{182}z_{240} - 4z_{188} - z_{188}^2 - 2z_{188}z_{194} - 2z_{188}z_{200} - 2z_{188}z_{207}$
	$-2z_{188}z_{213} - 2z_{188}z_{216} - 2z_{188}z_{217} - 2z_{188}z_{220} - 2z_{188}z_{226} - 2z_{188}z_{233}$
	$-2z_{188}z_{238} - 2z_{188}z_{240} - 4z_{194} - z_{194}^2 - 2z_{194}z_{200} - 2z_{194}z_{207}$
	$-2z_{194}z_{213} - 2z_{194}z_{216} - 2z_{194}z_{217} - 2z_{194}z_{220} - 2z_{194}z_{226} - 2z_{194}z_{233}$
	$-2z_{194}z_{238} - 2z_{194}z_{240} - 4z_{200} - z_{200}^2 - 2z_{200}z_{207} - 2z_{200}z_{213}$
	$-2z_{200}z_{216} - 2z_{200}z_{217} - 2z_{200}z_{220} - 2z_{200}z_{226} - 2z_{200}z_{233} - 2z_{200}z_{238}$
	$-2z_{200}z_{240} - 4z_{207} - z_{207}^2 - 2z_{207}z_{213} - 2z_{207}z_{216} - 2z_{207}z_{217}$
	$-2z_{207}z_{220} - 2z_{207}z_{226} - 2z_{207}z_{233} - 2z_{207}z_{238} - 2z_{207}z_{240} - 4z_{213}$
	$-z_{213}^2 - 2z_{213}z_{216} - 2z_{213}z_{217} - 2z_{213}z_{220} - 2z_{213}z_{226} - 2z_{213}z_{233}$
	$-2z_{213}z_{238} - 2z_{213}z_{240} - 4z_{216} - z_{216}^2 - 2z_{216}z_{217} - 2z_{216}z_{220}$
	$-2z_{216}z_{226} - 2z_{216}z_{233} - 2z_{216}z_{238} - 2z_{216}z_{240} - 4z_{217} - z_{217}^2$
	$-2z_{217}z_{220} - 2z_{217}z_{226} - 2z_{217}z_{233} - 2z_{217}z_{238} - 2z_{217}z_{240} - 4z_{220}$
	$-z_{220}^2 - 2z_{220}z_{226} - 2z_{220}z_{233} - 2z_{220}z_{238} - 2z_{220}z_{240} - 4z_{226}$
	$-z_{226}^2 - 2z_{226}z_{233} - 2z_{226}z_{238} - 2z_{226}z_{240} - 4z_{233} - z_{233}^2$
	$-2z_{233}z_{238} - 2z_{233}z_{240} - 4z_{238} - z_{238}^2 - 2z_{238}z_{240} - 4z_{240}$
	$-z_{240}^2$
$Q_{120} =$	$z_{131} - z_{139}z_{140} - z_{140}$
$Q_{121} =$	$1 - z_{123} - z_{124} - z_{130}z_{141} + z_{135}$
$Q_{122} =$	$-z_{129} - z_{130}z_{142} + z_{132}$
$Q_{123} =$	$-1 + z_{133} - 8z_{143} - 4z_{143}z_{145} - 4z_{143}z_{150} - 4z_{143}z_{160}$
	$-4z_{143}z_{164} - 4z_{143}z_{168} - 4z_{143}z_{175} - 4z_{143}z_{182} - 4z_{143}z_{188} - 4z_{143}z_{194}$
	$-4z_{143}z_{200} - 4z_{143}z_{207} - 4z_{143}z_{213} - 4z_{143}z_{216} - 4z_{143}z_{217} - 4z_{143}z_{220}$
	$-4z_{143}z_{226} - 4z_{143}z_{233} - 4z_{143}z_{238} - 4z_{143}z_{240} - z_{216} - z_{217}$

Residual	Integer polynomial (continued across rows when necessary)
	$-z_{240}$
$Q_{124} =$	$-z_{123}z_{145} - z_{124}z_{145} + z_{146}$
$Q_{125} =$	$-1 + z_{144}^2 + z_{145}^2 - z_{146}^2$
$Q_{126} =$	$-2 - 2z_3 - z_3^2 + 2z_3z_{123} + 2z_3z_{124} + 2z_{123}$ $+2z_{124} - z_{147}$
$Q_{127} =$	$z_3z_{145} - z_5 + z_{144} + z_{145} - z_{146} - z_{147}z_{148}$
$Q_{128} =$	$-z_{123}z_{150} - z_{124}z_{150} + z_{151}$
$Q_{129} =$	$-1 + z_{149}^2 + z_{150}^2 - z_{151}^2$
$Q_{130} =$	$-5 - 4z_4 - z_4^2 + 2z_4z_{123} + 2z_4z_{124} + 4z_{123}$ $+4z_{124} - z_{152}$
$Q_{131} =$	$z_4z_{150} - z_{21} + z_{149} + 2z_{150} - z_{151} - z_{152}z_{153}$
$Q_{132} =$	$-100 - 20z_{12} - z_{12}^2 - 2z_{12}z_{13} - 2z_{12}z_{14} - 2z_{12}z_{15}$ $-2z_{12}z_{16} - 2z_{12}z_{17} - 2z_{12}z_{18} - 2z_{12}z_{19} - 2z_{12}z_{20} - 20z_{13}$ $-z_{13}^2 - 2z_{13}z_{14} - 2z_{13}z_{15} - 2z_{13}z_{16} - 2z_{13}z_{17} - 2z_{13}z_{18}$ $-2z_{13}z_{19} - 2z_{13}z_{20} - 20z_{14} - z_{14}^2 - 2z_{14}z_{15} - 2z_{14}z_{16}$ $-2z_{14}z_{17} - 2z_{14}z_{18} - 2z_{14}z_{19} - 2z_{14}z_{20} - 20z_{15} - z_{15}^2$ $-2z_{15}z_{16} - 2z_{15}z_{17} - 2z_{15}z_{18} - 2z_{15}z_{19} - 2z_{15}z_{20} - 20z_{16}$ $-z_{16}^2 - 2z_{16}z_{17} - 2z_{16}z_{18} - 2z_{16}z_{19} - 2z_{16}z_{20} - 20z_{17}$ $-z_{17}^2 - 2z_{17}z_{18} - 2z_{17}z_{19} - 2z_{17}z_{20} - 20z_{18} - z_{18}^2$ $-2z_{18}z_{19} - 2z_{18}z_{20} - 20z_{19} - z_{19}^2 - 2z_{19}z_{20} - 20z_{20}$ $-z_{20}^2 + z_{155}$
$Q_{133} =$	$-1 + 2z_{12}z_{123} + 2z_{12}z_{124} + 2z_{13}z_{123} + 2z_{13}z_{124} + 2z_{14}z_{123}$ $+2z_{14}z_{124} + 2z_{15}z_{123} + 2z_{15}z_{124} + 2z_{16}z_{123} + 2z_{16}z_{124} + 2z_{17}z_{123}$ $+2z_{17}z_{124} + 2z_{18}z_{123} + 2z_{18}z_{124} + 2z_{19}z_{123} + 2z_{19}z_{124} + 2z_{20}z_{123}$ $+2z_{20}z_{124} + 20z_{123} + 20z_{124} - z_{154} - z_{155}$
$Q_{134} =$	$z_{12}z_{150} + z_{13}z_{150} + z_{14}z_{150} + z_{15}z_{150} + z_{16}z_{150} + z_{17}z_{150}$ $+z_{18}z_{150} + z_{19}z_{150} + z_{20}z_{150} - z_{119} + z_{149} + 10z_{150}$ $-z_{151} - z_{154}z_{156}$
$Q_{135} =$	$-10 - 6z_4 - z_4^2 + 2z_4z_{123} + 2z_4z_{124} - 2z_4z_{173}$ $+6z_{123} + 2z_{123}z_{173} + 6z_{124} + 2z_{124}z_{173} - z_{157} - 6z_{173}$ $-z_{173}^2$
$Q_{136} =$	$-1 + z_4z_{145} + z_{144} + 3z_{145} + z_{145}z_{173} - z_{146}$ $-z_{157}z_{158} - z_{171} - z_{172}$
$Q_{137} =$	$-z_{123}z_{160} - z_{124}z_{160} + z_{161}$
$Q_{138} =$	$-1 + z_{159}^2 + z_{160}^2 - z_{161}^2$
$Q_{139} =$	$-1 + z_4z_{160} - z_{157}z_{162} + z_{159} + 3z_{160} + z_{160}z_{173}$ $-z_{161} - z_{203} - z_{204}$
$Q_{140} =$	$-z_{123}z_{164} - z_{124}z_{164} + z_{165}$
$Q_{141} =$	$-1 + z_{163}^2 + z_{164}^2 - z_{165}^2$
$Q_{142} =$	$-1 + z_4z_{164} - z_{157}z_{166} + z_{163} + 3z_{164} + z_{164}z_{173}$ $-z_{165} - z_{216} - z_{217}$
$Q_{143} =$	$-z_{123}z_{168} - z_{124}z_{168} + z_{169}$
$Q_{144} =$	$-1 + z_{167}^2 + z_{168}^2 - z_{169}^2$
$Q_{145} =$	$-2 - z_4 + z_4z_{168} - z_4z_{171} - z_4z_{172} + z_{167}$ $+4z_{168} + z_{168}z_{173} - z_{169} - z_{170} - z_{170}z_{178} - z_{170}z_{179}$ $-3z_{171} - 2z_{172}$
$Q_{146} =$	$-z_{123}z_{175} - z_{124}z_{175} + z_{176}$
$Q_{147} =$	$-1 + z_{174}^2 + z_{175}^2 - z_{176}^2$

Residual	Integer polynomial (continued across rows when necessary)
$Q_{148} =$	$-z_{21} - z_{21}z_{178} - z_{21}z_{179} + z_{180}$
$Q_{149} =$	$z_4z_{175} + z_{173}z_{175} + z_{174} + 4z_{175} - z_{176} - z_{177}z_{180} - z_{178}$
$Q_{150} =$	$-z_{123}z_{182} - z_{124}z_{182} + z_{183}$
$Q_{151} =$	$-1 + z_{181}^2 + z_{182}^2 - z_{183}^2$
$Q_{152} =$	$z_4z_{182} + z_{173}z_{182} - z_{180}z_{184} + z_{181} + 4z_{182} - z_{183} - z_{185}$
$Q_{153} =$	$z_{178} + z_{179} - z_{185} - z_{186}$
$Q_{154} =$	$-z_{123}z_{188} - z_{124}z_{188} + z_{189}$
$Q_{155} =$	$-1 + z_{187}^2 + z_{188}^2 - z_{189}^2$
$Q_{156} =$	$z_4z_{188} + z_{173}z_{188} - z_{180}z_{190} + z_{187} + 4z_{188} - z_{189} - z_{191}$
$Q_{157} =$	$z_{178} + z_{179} - z_{191} - z_{192}$
$Q_{158} =$	$-z_{123}z_{194} - z_{124}z_{194} + z_{195}$
$Q_{159} =$	$-1 + z_{193}^2 + z_{194}^2 - z_{195}^2$
$Q_{160} =$	$z_4z_{194} + z_{173}z_{194} - z_{180}z_{196} + z_{193} + 4z_{194} - z_{195} - z_{197}$
$Q_{161} =$	$z_{178} + z_{179} - z_{197} - z_{198}$
$Q_{162} =$	$-z_{123}z_{200} - z_{124}z_{200} + z_{201}$
$Q_{163} =$	$-1 + z_{199}^2 + z_{200}^2 - z_{201}^2$
$Q_{164} =$	$-z_{21} - z_{21}z_{203} - z_{21}z_{204} + z_{205}$
$Q_{165} =$	$z_4z_{200} + z_{173}z_{200} + z_{199} + 4z_{200} - z_{201} - z_{202}z_{205} - z_{203}$
$Q_{166} =$	$-z_{123}z_{207} - z_{124}z_{207} + z_{208}$
$Q_{167} =$	$-1 + z_{206}^2 + z_{207}^2 - z_{208}^2$
$Q_{168} =$	$z_4z_{207} + z_{173}z_{207} - z_{205}z_{209} + z_{206} + 4z_{207} - z_{208} - z_{210}$
$Q_{169} =$	$z_{203} + z_{204} - z_{210} - z_{211}$
$Q_{170} =$	$-z_{123}z_{213} - z_{124}z_{213} + z_{214}$
$Q_{171} =$	$-1 + z_{212}^2 + z_{213}^2 - z_{214}^2$
$Q_{172} =$	$-z_{21} - z_{21}z_{216} - z_{21}z_{217} + z_{218}$
$Q_{173} =$	$z_4z_{213} + z_{173}z_{213} + z_{212} + 4z_{213} - z_{214} - z_{215}z_{218} - z_{216}$
$Q_{174} =$	$-z_{123}z_{220} - z_{124}z_{220} + z_{221}$
$Q_{175} =$	$-1 + z_{219}^2 + z_{220}^2 - z_{221}^2$
$Q_{176} =$	$z_4z_{220} + z_{173}z_{220} - z_{218}z_{222} + z_{219} + 4z_{220} - z_{221} - z_{223}$
$Q_{177} =$	$z_{216} + z_{217} - z_{223} - z_{224}$
$Q_{178} =$	$-z_{123}z_{226} - z_{124}z_{226} + z_{227}$
$Q_{179} =$	$-1 + z_{225}^2 + z_{226}^2 - z_{227}^2$
$Q_{180} =$	$-z_{21} - z_{21}z_{171} - z_{21}z_{172} + z_{231}$
$Q_{181} =$	$z_4z_{226} + z_{173}z_{226} + z_{225} + 4z_{226} - z_{227} - z_{228}z_{231} - z_{229}$

Residual	Integer polynomial (continued across rows when necessary)
$Q_{182} =$	$z_{171} + z_{172} - z_{229} - z_{230}$
$Q_{183} =$	$-z_{123}z_{233} - z_{124}z_{233} + z_{234}$
$Q_{184} =$	$-1 + z_{232}^2 + z_{233}^2 - z_{234}^2$
$Q_{185} =$	$z_4z_{233} + z_{173}z_{233} - z_{180}z_{235} + z_{232} + 4z_{233} - z_{234}$ $-z_{236}$
$Q_{186} =$	$z_{178} + z_{179} - z_{236} - z_{237}$
$Q_{187} =$	$1152 + 78z_0 + 3z_0^2 + 2z_0z_1 + 4z_0z_2 + 2z_0z_3$ $+2z_0z_9 - 2z_0z_{145} - 2z_0z_{150} - 2z_0z_{168} + 32z_1 + 2z_1^2$ $+4z_1z_2 + 2z_1z_9 - 2z_1z_{150} - 2z_1z_{164} + 64z_2 + 5z_2^2$ $+6z_2z_9 - 4z_2z_{150} - 2z_2z_{160} + 24z_3 + z_3^2 - 2z_3z_{168}$ $+34z_9 + 2z_9^2 - 2z_9z_{150} - 2z_9z_{160} + 52z_{12} + 2z_{12}^2$ $+26z_{12}z_{13} + 8z_{12}z_{23} + 4z_{12}z_{26} + 2z_{12}z_{30} + 2z_{12}z_{32} + 2z_{12}z_{34}$ $+2z_{12}z_{36} + 4z_{12}z_{40} + 8z_{12}z_{42} + 4z_{12}z_{43} + 2z_{12}z_{46} + 4z_{12}z_{68}$ $-2z_{12}z_{175} - 2z_{12}z_{233} + 652z_{13} + 170z_{13}^2 + 104z_{13}z_{23} + 52z_{13}z_{26}$ $+26z_{13}z_{30} + 26z_{13}z_{32} + 26z_{13}z_{34} + 26z_{13}z_{36} + 52z_{13}z_{40} + 104z_{13}z_{42}$ $+52z_{13}z_{43} + 26z_{13}z_{46} + 52z_{13}z_{68} - 2z_{13}z_{182} - 26z_{13}z_{233} + 2z_{14}$ $+z_{14}^2 - 2z_{14}z_{188} + 2z_{15} + z_{15}^2 - 2z_{15}z_{194} + 2z_{16}$ $+z_{16}^2 - 2z_{16}z_{200} + 2z_{17} + z_{17}^2 - 2z_{17}z_{207} + 2z_{18}$ $+z_{18}^2 - 2z_{18}z_{213} + 2z_{19} + z_{19}^2 - 2z_{19}z_{220} + 2z_{20}$ $+z_{20}^2 - 2z_{20}z_{226} + 200z_{23} + 16z_{23}^2 + 16z_{23}z_{26} + 8z_{23}z_{30}$ $+8z_{23}z_{32} + 8z_{23}z_{34} + 8z_{23}z_{36} + 16z_{23}z_{40} + 32z_{23}z_{42} + 16z_{23}z_{43}$ $+8z_{23}z_{46} + 16z_{23}z_{68} - 8z_{23}z_{233} + 100z_{26} + 4z_{26}^2 + 4z_{26}z_{30}$ $+4z_{26}z_{32} + 4z_{26}z_{34} + 4z_{26}z_{36} + 8z_{26}z_{40} + 16z_{26}z_{42} + 8z_{26}z_{43}$ $+4z_{26}z_{46} + 8z_{26}z_{68} - 4z_{26}z_{233} + 50z_{30} + z_{30}^2 + 2z_{30}z_{32}$ $+2z_{30}z_{34} + 2z_{30}z_{36} + 4z_{30}z_{40} + 8z_{30}z_{42} + 4z_{30}z_{43} + 2z_{30}z_{46}$ $+4z_{30}z_{68} - 2z_{30}z_{233} + 50z_{32} + z_{32}^2 + 2z_{32}z_{34} + 2z_{32}z_{36}$ $+4z_{32}z_{40} + 8z_{32}z_{42} + 4z_{32}z_{43} + 2z_{32}z_{46} + 4z_{32}z_{68} - 2z_{32}z_{233}$ $+50z_{34} + z_{34}^2 + 2z_{34}z_{36} + 4z_{34}z_{40} + 8z_{34}z_{42} + 4z_{34}z_{43}$ $+2z_{34}z_{46} + 4z_{34}z_{68} - 2z_{34}z_{233} + 50z_{36} + z_{36}^2 + 4z_{36}z_{40}$ $+8z_{36}z_{42} + 4z_{36}z_{43} + 2z_{36}z_{46} + 4z_{36}z_{68} - 2z_{36}z_{233} + 100z_{40}$ $+4z_{40}^2 + 16z_{40}z_{42} + 8z_{40}z_{43} + 4z_{40}z_{46} + 8z_{40}z_{68} - 4z_{40}z_{233}$ $+200z_{42} + 16z_{42}^2 + 16z_{42}z_{43} + 8z_{42}z_{46} + 16z_{42}z_{68} - 8z_{42}z_{233}$ $+100z_{43} + 4z_{43}^2 + 4z_{43}z_{46} + 8z_{43}z_{68} - 4z_{43}z_{233} + 50z_{46}$ $+z_{46}^2 + 4z_{46}z_{68} - 2z_{46}z_{233} + 100z_{68} + 4z_{68}^2 - 4z_{68}z_{233}$ $-z_{123}z_{239} - z_{124}z_{239} - 24z_{145} + z_{145}^2 - 30z_{150} + z_{150}^2$ $-4z_{160} + z_{160}^2 - 2z_{164} + z_{164}^2 - 24z_{168} + z_{168}^2$ $-2z_{175} + z_{175}^2 - 2z_{182} + z_{182}^2 - 2z_{188} + z_{188}^2$ $-2z_{194} + z_{194}^2 - 2z_{200} + z_{200}^2 - 2z_{207} + z_{207}^2$ $-2z_{213} + z_{213}^2 - 2z_{220} + z_{220}^2 - 2z_{226} + z_{226}^2$ $-50z_{233} + z_{233}^2 + z_{239}$
$Q_{188} =$	$-96 - 3z_0 - 2z_1 - 3z_2 - 2z_3 - 2z_4$ $-z_5 - 2z_9 - 3z_{12} - 15z_{13} - 2z_{14} - 2z_{15}$ $-2z_{16} - 2z_{17} - 2z_{18} - 2z_{19} - 2z_{20} - z_{21}$ $-4z_{23} - 2z_{26} - z_{30} - z_{32} - z_{34} - z_{36}$ $-2z_{40} - 4z_{42} - 2z_{43} - z_{46} - 2z_{68} - z_{119}$ $-z_{171} - z_{172} - z_{173} - z_{203} - z_{204} + z_{240}$

B Coordinate meanings

These names identify compiler coordinates; they impose no extra conditions on the variables. The preceding residual list is the complete arithmetic definition. Names of eliminated variables occur only in the supplied restoration certificates.

Index	Name	Index	Name
0	t0	121	mul.426
1	x	122	mul.427
2	fact0.output	123	shared.lower_A
3	mul.1	124	shared.A_slack
4	reciprocal.Bminus2	125	shared.Zminus1
5	pow0.output	126	mul.431
6	fact0.remainder	127	mul.432
7	fact0.slack	128	modular.root_quotient
8	endpoint.u0	129	index_cap.X
9	endpoint.r0	130	index_cap.U
10	endpoint.ut	131	index_cap.V
11	mul.12	132	index_cap.S
12	crt.encoded.a	133	index_cap.R
13	crt.encoded.b	134	index_cap.q4minus1
14	crt.encoded.c	135	mul.441
15	crt.encoded.d	136	mul.443
16	crt.encoded.S	137	mul.444
17	crt.encoded.Sp	138	mul.445
18	crt.encoded.u	139	index_cap.Vquotient_minus1
19	crt.encoded.vq	140	mul.446
20	crt.encoded_index_minus_one	141	index_cap.Bquotient
21	pow1.output	142	index_cap.Squotient
22	crt.index_quotient	143	index_cap.Rquotient
23	mul.41	144	index0.X
24	mul.48	145	index0.Y
25	mul.53	146	mul.455
26	mul.60	147	base0.T
27	mul.65	148	pow0.quotient
28	mul.79	149	index1.X
29	add.89	150	index1.Y
30	add.94	151	mul.465
31	add.106	152	base1.T
32	add.111	153	pow1.quotient
33	add.126	154	base2.T
34	add.131	155	mul.476
35	add.140	156	pow2.quotient
36	add.145	157	base3.T
37	mul.148	158	pow3.quotient
38	mul.153	159	index2.X
39	mul.160	160	index2.Y
40	mul.165	161	mul.493
41	add.176	162	pow4.quotient
42	add.181	163	index3.X
43	mul.184	164	index3.Y
44	mul.190	165	mul.499
45	add.204	166	pow5.quotient
46	add.209	167	index4.X
47	add.210	168	index4.Y
48	mul.219	169	mul.505
49	mul.224	170	choose0.modular_quotient
50	mul.230	171	choose0.modular_remainder
51	mul.237	172	choose0.modular_rslack
52	mul.242	173	choose0.modular_oslack
53	mul.248	174	index5.X
54	add.253	175	index5.Y
55	add.258	176	mul.517
56	mul.261	177	crt.range.a.modular_quotient
57	mul.268	178	crt.range.a.modular_remainder
58	mul.273	179	crt.range.a.modular_rslack

Index	Name	Index	Name
59	mul.279	180	mul.518
60	mul.286	181	index6.X
61	mul.291	182	index6.Y
62	mul.304	183	mul.526
63	add.307	184	crt.range.b.modular_quotient
64	mul.309	185	crt.range.b.modular_remainder
65	mul.322	186	crt.range.b.modular_rslack
66	add.332	187	index7.X
67	add.337	188	index7.Y
68	mul.340	189	mul.534
69	mul.343	190	crt.range.c.modular_quotient
70	sqd.0	191	crt.range.c.modular_remainder
71	mul.344	192	crt.range.c.modular_rslack
72	sqd.1	193	index8.X
73	mul.346	194	index8.Y
74	sqd.2	195	mul.542
75	mul.348	196	crt.range.d.modular_quotient
76	sqd.3	197	crt.range.d.modular_remainder
77	mul.350	198	crt.range.d.modular_rslack
78	sqd.4	199	index9.X
79	mul.352	200	index9.Y
80	sqd.5	201	mul.550
81	mul.354	202	crt.range.S.modular_quotient
82	sqd.6	203	crt.range.S.modular_remainder
83	mul.356	204	crt.range.S.modular_rslack
84	sqd.7	205	mul.551
85	mul.358	206	index10.X
86	sqd.8	207	index10.Y
87	sqd.9	208	mul.559
88	sqd.10	209	crt.range.Sp.modular_quotient
89	mul.360	210	crt.range.Sp.modular_remainder
90	mul.362	211	crt.range.Sp.modular_rslack
91	sqd.11	212	index11.X
92	mul.363	213	index11.Y
93	sqd.12	214	mul.567
94	mul.366	215	crt.range.u.modular_quotient
95	mul.368	216	crt.range.u.modular_remainder
96	mul.369	217	crt.range.u.modular_rslack
97	mul.371	218	mul.568
98	mul.373	219	index12.X
99	mul.374	220	index12.Y
100	sqd.13	221	mul.576
101	mul.375	222	crt.range.vq.modular_quotient
102	mul.377	223	crt.range.vq.modular_remainder
103	mul.378	224	crt.range.vq.modular_rslack
104	sqd.14	225	index13.X
105	mul.379	226	index13.Y
106	mul.380	227	mul.584
107	mul.381	228	crt.range.index.modular_quotient
108	mul.382	229	crt.range.index.modular_remainder
109	mul.383	230	crt.range.index.modular_rslack
110	mul.384	231	mul.585
111	mul.385	232	index14.X
112	mul.386	233	index14.Y
113	mul.387	234	mul.593
114	mul.388	235	crt.range.derived_label_sum.modular_quotient
115	mul.391	236	crt.range.derived_label_sum.modular_remainder

Index	Name	Index	Name
116	`mul.393`	237	`crt.range.derived_label_sum.modular_rslack`
117	`mul.396`	238	`index_cap.aggregate_slack`
118	`crt.nonnegative_quotient`	239	`index_cap.error_square_quotient`
119	`pow2.output`	240	`add.639`
120	`mul.425`		

C Reproduction and dependency map

The top-level `README.md` gives the selected files and exact replay command. The direct subdirectory contains the frozen compiler inputs, generator, whole expanded quartics, and elimination certificates. The positive subdirectory contains its fixed coefficient payloads, source equations, literal schedules, ordinary proof, and independent arithmetic checks. Retained upstream licenses and notices apply to their corresponding files.

The release verifier checks the file manifest, reproduces the selected mathematical artifacts, checks the complete principal residual appendix, and writes its receipt to a separate location. Source and artifact hashes pin the comparison and prevent an accidental mixture of constants or compiler versions. Hashes authenticate finite bytes; they do not replace the unbounded mathematical arguments.

References

[1] ProveIt Contributors, *ProveIt*, snapshot `f4d8db99e060339f3306a04d507c35bc1d0d68c5`, 24 October 2026. https://github.com/VladimirReshetnikov/ProveIt.

[2] ProveIt incoming research packet, *Shared Binomial Certificates and Short Carriers for Con(ZFC)*, 24 October 2026. Archive `ConZFC_Shared_Binomial_Certificates_2026-10-24.zip`, `docs/incoming` at the pinned snapshot. Its direct ordinary proof and finite source are retained in this packet.

[3] ProveIt incoming research packet, *One Pell Bound and One Error Congruence*, 24 October 2026. Archive `ConZFC_Shared_Pell_Bound_261_2026-10-24.zip`, same pinned directory. The one-anchor theorem is retained as `CAP_THEOREM.md`.

[4] ProveIt Contributors, *Coprime Fusion with the Auxiliary-Index Gap Majorant*, `Computability/DiophantineConsistency/candidates/ten-positive-gap-fusion/PROOF.md`, same pinned snapshot.

[5] ProveIt incoming research packet, *Round 15 Unified Binomial and Even Pell Index Equations*, 24 October 2026. Archive `ConZFC_Unified_Binomial_Even_Indices_Round15_2026-10-24.zip`, same pinned directory. The raw ceiling bound is retained in `direct/source/UNIFIED_PRODUCT_PROOF.md`.

[6] J. Bayer and M. David, *A Formal Proof of Complexity Bounds on Diophantine Equations*, ITP 2025, LIPIcs 352, Article 3, 2025. https://doi.org/10.4230/LIPIcs.ITP.2025.3.

[7] J. Bayer, M. David, M. Haßler, Y. Matiyasevich, and D. Schleicher, *Diophantine Equations over $\mathbb{Z}$: Universal Bounds and Parallel Formalization*, arXiv:2506.20909v3, 22 October 2026. https://arxiv.org/abs/2506.20909.